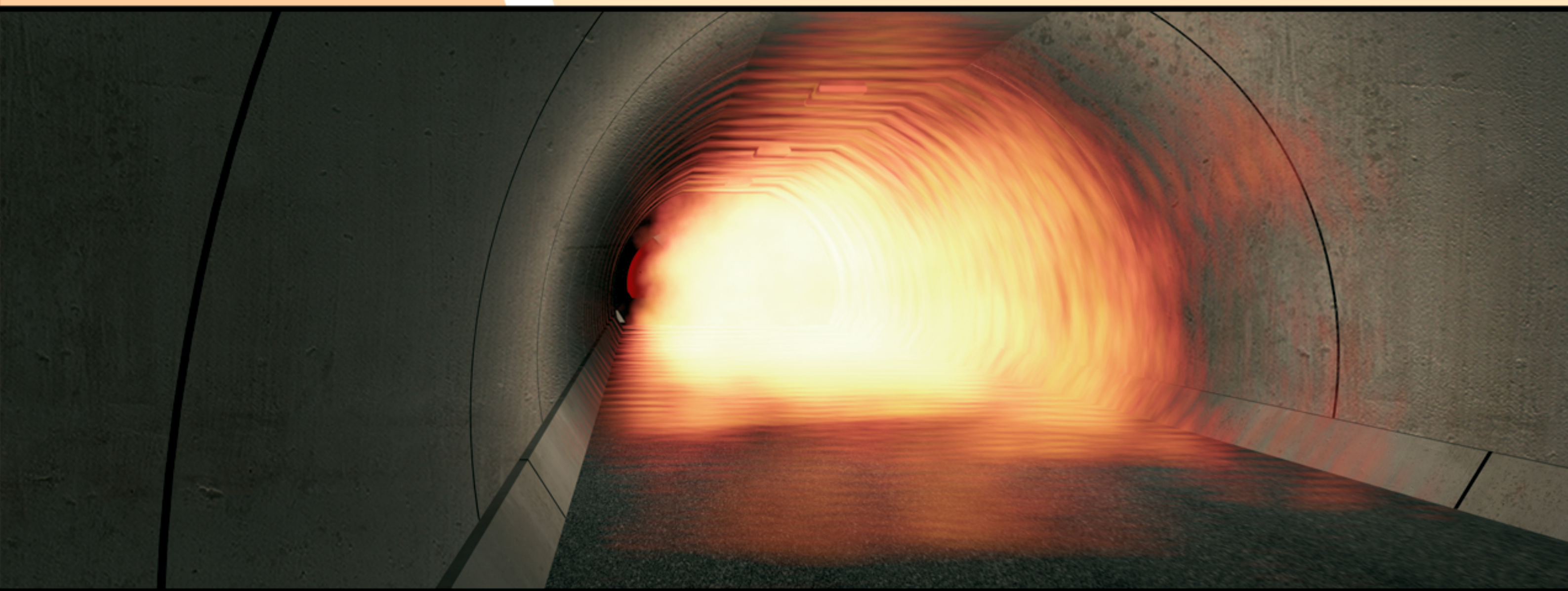
European Thematic Network Fire in Tunnels

Fire In Tunnels

Thematic network



Technical report part 2

Fire Safe Design - Rail Tunnels

supported by

Technical Report – Part 2

Fire Safe Design – rail tunnels

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Rail Tunnels

In accordance with the description of work package 3.1 and discussions at the network meetings, this section of the report covers:

- Structural safety facilities
- Safety equipment
- Reaction/resistance to fire

In addition some typical provisions on the following topics are also showed:
General design characteristics of railway tunnels
Emergency management
The work in WP3 includes both a listing of relevant guidelines and comparisons between those selected.
(The contents of the present document will be incorporated into the WP3 part of the FIT report. The section numbers will be subsections to the section 3.4 Rail Tunnels. For practical reasons the two first digits in the section numbers have been left out in this document.)

1 LIST OF COLLECTED GUIDELINES

The first part of the activity concerns the listing of relevant guidelines. It has been agreed that this should include regulations, guidelines, standards, and to some degree current best practices. Guidelines concerning construction are beyond the scope. Guidelines include relevant documents from European and international organisations and European countries, supplemented when relevant with guidelines from other important rail tunnel countries.

Based on available documents, specific searches and reference to similar work by UIC and UN/ECE, a number of relevant documents have been identified.
In addition a state-of-the-art report from USA is included together with reports from the task forces of UN/ECE and UIC.

The following countries are the main "rail-tunnel-countries" in Europe, i.e. those countries whose railway networks include by the highest total length of tunnels over 1000m long: Italy (608 km), Switzerland (298km), Germany (274 km), France (197 km), Norway (126 km), Austria (89 km), UK (90 km) and Spain (79 km).
The selection of guidelines has also been based on an evaluation, made by the authorities of each European country, concerning validity and sufficiency of their standards. According to the review made by UN/ECE the guidelines of the Netherlands and Sweden are also worthy of inclusion. The US standard is included as an important international reference.

The first level of reporting is a list of the documents including:
- Title, reference and date of the document
- The administrative status of the document in the country concerned

The second level includes
- An analytical summary in English of the contents stating the essential items beneficial for the topics compiled by WP3.
- The table of contents translated into English (as shown in the appendices)

1.1 *Table of references (national# Guidelines)*

Country	Title / Issued by	Reference	Date	Administrative status	Comments
Italy	• Linee guida per il miglioramento della sicurezza nelle gallerie ferroviarie", 25 Luglio 1997 • Ministry of the Interior, FS S.p.A., National Fire Brigade Corp.		Jul. 1997	Guideline for Italian Railway Network RFI issued by the standardization working party	-This guideline is the result of a working party with F.S. S.p.A, Ministry of the Interior and Fire Brigade Corp. **** Specific methodology for risk assessment and risk management for tunnel is covered by national regulations. - This document applies only to FS-RFI
	[2] Criteri progettuali per la realizzazione di piazzali di emergenza, le strade di accesso e le aree di atterraggio degli elicotteri ai fini della sicurezza delle gallerie ferroviarie (Agosto 98) **FS – RFI**		Aug. 1998	Technical Specification	- These documents apply only to FS-RFI
	[3] "Criteri progettuali per la realizzazione degli impianti idrico antincendio, elettrico e illuminazione, telecomunicazione, supervisione" (Aprile 2000) **FS – RFI**		Apr. 2000	Technical Specification	
	"Linee guida per la realizzazione del piano generale di emergenza per lunghe gallerie ferroviarie" (Ottobre 98) **FS – RFI**		Oct. 1998	Guideline	
	"Linee guida per l'elaborazione del piano interno di emergenza" (Giugno 2000) **FS-RFI**		Jun. 2000	Guideline	
	Linee guida per il tracciamento e la posa in opera di sistemi di supporto per cavo radiante nelle gallerie ferroviarie" (Aprile 01) **FS-RFI**		Apr. 2001	Guideline	

	RFI circular n° di/a1007/p/01/000562 del 04.06.2001: “Piano interno di emergenza per gallerie di lunghezza compresa tra i 5000m e 3000m” **FS-RFI**	Nota RFI n°di/a1007/p/01/000562	Jun. 2001	Circular	-These circulars constitute a further improvement to the 1997 Guidelines.
	RFI circular n° rfi/tdr/a1007/p/01/000512 del 17.12.2001: “Standard di sicurezza per nuove gallerie ferroviarie” **FS-RFI**	Nota RFI n°rfi/tdr/a1007/p/01/000512	Dec. 2001	Circular	
Switzerland	• Raccomandazione comune delle autorità di vigilanza sulle ferrovie della Germania, dell’Austria e della Svizzera in merito alla sicurezza dei viaggiatori in gallerie ferroviarie molto lunghe, 24.09.1992 • Swiss Federal office of Transport		Sept. 1992	Recommendation	**** Specific methodology for risk assessment and risk management for tunnels is covered by national regulations specific methodology developed by Swiss Federal Railways. The requisites set out therein are not deemed sufficient by the Swiss Authorities. The present shortcomings have been remedied in a draft
	[2] Prescrizioni svizzere sulla circolazione dei treni (14.12.2003) UFT Swiss Federal office of Transport	PCT(R300.1-.15),	Dec. 2003	Regulation	
	[3] Weisung I-AM-EB-31/00: Sichereith in bestehenden Tunnels, Infrastrukturmassnahmen zur Erleichterung der Selbstrettung , 06.12.2000. • Swiss Federal Railways	Weisung I-AM-EB-31/00	Dec. 2000	Technical specification	

	[4] Ausfuhrungsbestimmungen zur Eisenbahnverordvom 16 nung Oktober 2002	AB-EBV SR 742.141.11	Oct. 2002	Law	Swiss standard SIA 198 e SIA 198/1. In particular, safety in the Gotthard (57 km) and Lotschberg (34 km) tunnels was evaluated and implemented by the authorities on the basis of a special project called Projektorganisation Sicherheitsbericht Alp Transit.
	[5] Verordnung vom 27 Februar 1991 über den Schutz vor Störfällen **Störfällverordnungen StFV, SR 814.012**	StFV SR 814.01	Feb. 2001	Regulation	
	[6] Verordnung vom 23 November 1983 über Bau und Betrieb der Eisenbahnen **Eisenbahnverordnung EVB**	EVB SR 742.141.1	Nov. 2003	Regulation	

Germany	[1] Richtlinie Anforderung des Brand- und Katastrophenschutzes an den Bau und Betrieb von Eisenbahntunneln (01.07.1997 with ammendments from 30.07.1999) Eisenbahn-Bundesamt (EBA)		Jul. 1997	Guideline	
	[2] Leitfaden für den Brandschutz in Personenverkehrsanlagen der Eisenbahnen des Bundes und der Magnetschnellbahn Jan. 2001 **Eisenbahn-Bundesamt (EBA)**		Jan. 2001		
	[3] Anforderungen der DB Station&Service AG an den Brandschutz in Personenverkehrsanlagen Draft Version 15.03.2001 **Deutsche Bahn AG (DB)**		Mar. 2001	Guideline	
	[4] Fire and Disaster protection in railway tunnels **Deutsche Bahn AG (DB)**		Mar. 2004	Guideline	
France	[1] Instruction technique interministérielle relative à la sécurité dans les tunnels ferroviaires (08.07.1998) **Ministère de l'Equipement, des Transportes et du Logement**	IT N°98-300	Jul. 1998	Ministerial instruction	**** Public security services aim at reinforcing measures. **** Specific methodology for risk assessment and risk management for tunnels is covered by national regulations
	[2] Décret relatif a la sécurité' du réseau ferre national, 30.03.2000 **Ministère de l'Equipement, des Transportes et du Logement**		Mar. 2000	Legislative Decree	

Norway	[1] Substructure, regulations for new lines, tunnels – safety requirements, (01.01.2000) **Jernbaneverket**	Doc.nr. JD 520	Gen. 2000	Regulation	This applies to new tunnels only. Further refinements are needed in case of special conditions. **** Specific methodology for risk assessment and risk management for tunnel is considered by Norwegian council for building standardization. - This document applies only to Jernbaneverket
Austria	[1] Richtlinie Bau und Betrieb von neuen Eisenbahntunneln bei Haupt- und Nebenbahnen, Anforderungen des Brand- und Katastrophenschutzes (1. Ausgabe 2000) **Österreichischer Bundesfeuerwehrverband**	ÖBFV-RL A-12	Ed. 2000	Guideline supported by the Federal Fire Brigade association.	**** Texts are considered sufficient to ensure safety, but need harmonization. **** Specific methodology for risk assessment and risk management for tunnels is covered by national regulations (EB and Partner).
	[2] Eisbav: Eisenbahn-ArbeitnehmerInnershutzverordnung **Verkehrs-Arbeitsinspektorat**	BGBI n° 364/1999, n° 444/1999	Jun. 2001	Law	
	[3] Allgemeines Sicherheitskonzept für mittlere Tunnel **OBB - Austrian Federal Ministry of Transport - Eisenbahn – Hochleistungsstrecken AG**	Doc n° 94086-4 dec. 1995 RB/PJ/CF/HF	Dec 1995	Technical Addendum	
	[4] Allgemeines Sicherheitskonzept für mittlere Bestandestunnel **OBB- Austrian Federal Ministry of Transport - Eisenbahn – Hochleistungsstrecken AG**	Doc n° 94086-20 Aug. 1996 GLA/CF/PJ	Aug. 1996	Technical Addendum	

	[5] Richtilinien für das Entwerfen von Bahnanlagen - Hochleistungsstrcken (HL- Richtilinien)			Guideline	
	[6] Handbuch „Feuerwehreinsatz im Gleisbereich“ **ÖBB - ÖBFV**			Manual	
UK	«Railway Safety Principles and Guidance, Part 2, Section A, Guidance on the infrastructure». Chapter 5 : TUNNELS **UK Health and Safety Executive**	Part 2, Section A ISBN 0-7176-1732-7	Mar. 2002	Regulation issued by the Health and Safety Executive.	This guidance is not compulsory and one is free to take any other action that complies with the (risk assessment based) safety laws. However, compliance with the guidance is taken to give compliance with all legal requirements. **** Specific methodology for risk assessment and risk management for tunnels is covered by national regulations. All operators must submit and have approved a Railway Safety Case before they are allowed to operate.
	System Safety Requirements for New and Re-opened Tunnels **Railway Safety**	RGS GC/RT 5114 Draft 3f	Dec 2002	Guideline	Railway Group Guidance Notes are non-mandatory documents providing helpful information relating to the

	Guidance on System Safety Requirements in New and Re-opened Tunnels **Railway Safety**	RGG GC/GN 5614 Draft 2f	Dec 2002	Guideline	control of hazards and often set out a suggested approach, which may be appropriate for Railway Group members to follow

Spain	• Instruccion obras subterraneas (01.12.1998) • RENFE - Ministirio de Fomento	BOE n°287	Dec. 1998	Regulation	**** Existing text are considered partially sufficient to ensure safety in tunnels. **** Specific methodology for risk assessment and risk management for tunnels is covered by national regulations.
	[2] Medidas de seguridad en nuevos tuneles ferroviarios (04.2000) • RENFE		Apr. 2000	Guideline	
The Netherlands	The Dutch Vision on Safety in Road and Rail Tunnels (Draft) **Ministries of transports and of Inland Affairs**	UN/ECE Inland Transport Committee informal Doc. N°1	2003		**** Specific methodology for risk assessment and risk management for tunnels is covered by national regulations: quantitative risk assessment and deterministic or scenario analysis.
Sweden	Säkerhet i järnnvägstunnlar, Ambitionsniva och värderingsmethodik, Handbok BVH 585.30 (01.09.1997) Zusammenfassung auf Englisch Banverket	Handbok BVH 585.30	Sept. 1997	Guideline	
Finland	• Technical Regulation and Guidelines for Railways" "Railway Tunnels" • RAMO, Finnish Rail Administration Board	RAMO section n°18	Oct. 2002	Technical Regulation	
USA	• NFPA 130 Standards for Fixed Guideway Transit and Passenger Rail Systems • National Fire Association	NFPA 130	May. 2003	American National Standard, issued by Standards Council.	***

Not all the guidelines can be considered truly national in scope (e.g. Italian guidelines are valid only for FS-RFI, the Norwegian for Jernbaneverket, etc)
**** Comments by Governmental representatives (National Road Administrations or Traffic Ministries) to the questionnaire issued by UN/ECE in 2000 to the question: "(Are there any legislation, regulations, recommendations on safety in road tunnels in your country and) do you consider the above texts sufficient?"
*** USA was not included in the UN/ECE review

1.2 Table of references (other reference documents)

Organisation	Title	Reference	Date	Administrative status	Comments
UIC International Union of Railways	Safety in Railway Tunnels -Recommendations for Safety Measures Final Report.	Fiche 779-9	Sept. 2002	Compendium of safety measures. Guideline.	
	Measures to limit and reduce the risk of accidents in underground railway installations with particular reference to the risk of fire and the transport of dangerous goods.	UIC Report IF 4/91	June 1991	Guideline	
UN United Nations	Report of the ad hoc multidisciplinary group of experts on safety in tunnels (Rail).	Trans/ac .9/8	30 July 2003	Recommendations	
	Questionnaire on safety in railway tunnels Replies by different Countries. (France – Norway – UK)	UN Inf. Doc. 14-15-16	2002	Replies to questionnaire	
	A summary of accidents in railway tunnels Department Of Transport, UK	UN Inf. Doc N°8	Sept. 2002	Report	
	Railways Tunnels in Europe and North America. UN Secretariat	UN Inf. Doc N°7	May 2002	Note	
	Vehicles fire and fire safety in tunnels Centre For Fire Safety In Transport, UK	UN Inf. Doc N° 9	Sept. 2002	ARTICLE	
	Peut-on garantir la sécurité des voyageurs dans les longs tunnels ferroviaires? ALPTRANSIT GOTHARD SA	UN Inf. Doc N°1	Jun. 2003		
	The Safety of the Swiss railway tunnels analysis of the federal office of transport UN-Secretariat	UN Inf. Doc N°2	Jun. 2003	Summary Report	
	Protection against fire and other catastrophic events in railways tunnels. Deutsche Bahn AG	UN Inf. Doc N°12	Feb. 2002		
	[9] Fire protection in transport tunnels (Germany)	UN Inf. Doc N° 13	Jun. 2003	Summary Report	

❖ List of collected guidelines

Organisation	Title	Reference	Date	Administrative status	Comments
	[10] Report of the ad hoc multidisciplinary group of experts on safety in tunnels (Rail).	Trans/ac .9/9	Dec 2003	Recommendations	
AEIF European Association for Railway Interoperability	Mandate for CR second priority TSIs (version 04)		Jun. 2002		
SBB-CFF-FSS Swiss Railway	Nutzungsanforderungen an neue Eisenbahntunnel		Mar. 2001	Recommendation	
ALPETUNNEL GEIGE	Safety Criteria for Operation, Lyon-Turin, base tunnel	A96074/r01-D/RR/NR	Apr. 1997	Summary of guidelines	
RFF Réseau Ferré de France	Safety principles for new long tunnels for freight and the rail motorway.	Reseau Ferr. 25/03/2003	Mar. 2003	Note	
	Official Policy for making existing tunnels safer during renovation and/or widening work on the National Railway Network.	Dir.Reseau Ferr. 28/03/2003	Mar. 2003	Note	
DB Deutsche Bahn	Fire Disaster Control on the Cologne-Frankfurt new build line		Dec. 2002	Safety Concepts	
BBT Brenner Basis Tunnel	Sicurezza nelle grandi gallerie di base alpine Brenner Basis Tunnel		Oct. 2001	Safety Concepts	
Danish National Railway Agency - Denmark	The Great Belt Link, Banestyrelsen –			Safety Concepts	
The Netherlands	Integraal Veiligheidsplan HSL-Zuid HSL-Zuid		2001	Safety Concepts	
	Memo Brandcurve HSL-Zuid HSL-Zuid		2001	Safety Concepts	

Organisation	Title	Reference	Date	Administrative status	Comments
	Beveiligingsconcept HSL-Zuid, Teile A, B1, B2, B3 HSL-Zuid		2001	Safety Concepts	
	High speed line south: safety concept- Green Heart Tunnel Government of Netherlands	UN-Informal Doc N°10	Sept. 2002	Safety Concepts	
OFT Swiss Federal Office of Transport	Rapport final sur la Sécurité dan le tunnels ferroviaires suisses		Jan. 2001	Technical report	

1.3 Analytical summaries (national# guidelines)

1.3.1 Italy

1.3.1.1 Linee guida per il miglioramento della sicurezza nelle gallerie ferroviarie", 25 July 1997

Ministry of the Interior, FS S.p.A., National Fire Brigade Corp.
Summary:
These guidelines were defined within the framework of a mixed working party (F.S. S.p.A. – Ministry of the Interior, C.N. V.V.F.) and came into force in July 1997. The related document does not have the force of law, although it summarises the recommendation of the working party. The document identifies appropriate safety measures to be observed during the design and management stages prior to the construction of new infrastructure and the upgrading of existing works. These are designed to safeguard the wellbeing of persons (passengers and rescuers) with regard to accident risks in tunnels and especially in the event of fire. The document is divided into three chapters dealing, respectively, with existing tunnels (chapter 1), tunnels under construction (chapter II) and tunnels yet to be constructed (chapter III). The areas of application of the document up to 1997 are: Mixed traffic lines; Tunnels of between 5 and 20 km in length. In major base tunnels (L> 20 km) the right to make use of specific studies is guaranteed. The document does not consider either station tunnels or metro tunnels as these are subject to specific regulations.

1.3.1.2 Criteri progettuali per la realizzazione di piazzali di emergenza, le strade di accesso e le aree di atterraggio degli elicotteri ai fini della sicurezza delle gallerie ferroviarie (Agosto 98)

FS – RFI
Summary:
This document indicates the design criteria for emergency areas used for stationing rescue vehicles, and for access roads and helidecks, related to railway tunnels of between 5 and 20 km, as indicated in the Guidelines for improving safety in railway tunnels [1.3.1.1].

1.3.1.3 Linee guida per la realizzazione del piano generale di emergenza per lunghe gallerie ferroviarie (Ottobre 98)

FS – RFI
Summary:
This document supplements the Guidelines by indicating about the definition and readiness of specific emergency plans to be used in order to deal with every possible and foreseeable accident scenario concerning so-called residual risks. The document sets out to provide an objective, clear and practical instrument, that is applicable in every situation and risk scenario. The document indicates the fundamental criteria and the reference contents for the design and execution of an emergency plan whose preparation should, in any case, take the form of an omni-comprehensive edition within the framework of a programme designed to keep all risks to a minimum by the competent and responsible authorities (Decree Law 626/94).

1.3.1.4 Criteri progettuali per la realizzazione degli impianti idrico antincendio, elettrico e illuminazione, telecomunicazione, supervisione (Aprile 2000)

FS – RFI
Summary:
This document represents a primer and a guide for compiling the project and has been drawn up on the basis of the requirements of the Guidelines. The instructions set forth in the document must be supplemented and adapted in relation to the real characteristics of the tunnel to be fitted out, and must take account of the condition of the places. Although the subject matter of the document concerns the first two chapters of the Guidelines [1.3.1.1], referring to existing tunnels and those under construction, it also represents an obligatory reference for future works in terms of the basic technical choices of the plant and equipment to be used.

1.3.1.5 Linee guida per l'elaborazione del piano interno di emergenza (Giugno 2000)

FS-RFI
Summary:
This document is a guide for the preparation of an Internal Emergency Plan in long railway tunnels. The document is concerned with ways of dealing with an accident in the shortest possible timescale while limiting the inconvenience to the persons involved, preventing the spread of the damage and guaranteeing the continuance of normal operations in total safety. The document was drawn up for rescue operations in railway tunnels, whatever the timetable for work to upgrade tunnel safety. It proposes the planning and coordination of all operational aspects of alarm and intervention of personnel for events that affect movement within a tunnel, even when the intervention of non-FS rescue agencies is requested, as provided for by the Guidelines. Example Emergency-action flow-charts are provided for each accident scenario, which constitute a guide for the compilation of the plan and are to be supplemented wherever necessary.

1.3.1.6 "Linee guida per il tracciamento e la posa in opera di sistemi di supporto per cavo radiante nelle gallerie ferroviarie" (Aprile 01)

FS – RFI
Summary:
These instructions give model solutions and the guidelines to be followed in designing and building support systems for the 'leaky' cables used to promulgate external communication network signals inside railway tunnels. In addition, model solutions to be adopted are laid down regarding the tracing, boring and laying of ducting for power cables and optical fibre feeding radio propagation plant in the tunnel, when special culverts or conduits are not available or when no space is available in them. The purpose of the instructions is to make such activities possible in parallel with retrofitting or upgrading work on the power lines or reinforcement in railway tunnels. The tracing, boring or laying operations to be conducted in order to implement the various plant solutions are described, with reference to the type of cables to be successively laid, the type of support, the type of railway line, the type of suspension to be used and the type of tunnel.

1.3.1.7 RFI circular n° di/a1007/p/01/000562 del 04.06.2001: “Piano interno di emergenza per gallerie di lunghezza compresa tra i 5.000m e 3.000m

FS-RFI
Summary:
This circular makes a provision that Internal Emergency Plans shall be drawn up for tunnels of between 3 and 5 km in length, therefore extending the requirement to all tunnels whose length is >3 km.

1.3.1.8 RFI circular n° rfi/tdr/a1007/p/01/000512 del 17.12.2001: “Standard di sicurezza per nuove gallerie ferroviarie”

FS-RFI
Summary:
This circular confirms model solutions for the most appropriate cross-sections for new tunnels (single or double bore) on the basis of their length in order to adopt a design standard that will ensure a level of safety in line with that of the other European networks, also taking into account the high operating speeds used and the mixed traffic involved. Furthermore, the circular, lays down the infrastructure features which should be included to enhance safety: bypasses, refuges, signalling plant that prevents other trains from arriving once the alarm is given.

1.3.2 Switzerland

General comments:
Safety in Swiss railway tunnels is based on various special purpose regulations, especially AB-EBV, SR 742.141.1, the law on railway infrastructure and the recommendations of the Federal Transport Office. The requisites set out in these regulations are no longer deemed sufficient by the Swiss authorities, especially the law AB-EBV, that will be changed appropriately. Moreover, the shortcomings have been remedied in the drafts of the Swiss standard SIA 198 and SIA 198/1. Safety in the Gotthard (57 km) and Lotschberg (34 km) tunnels was evaluated and implemented by the authorities through a special project called Projektorganisation Sicherheitsbericht ALP TRANSIT. Specific handbooks issued by the Swiss Federal Railways (SBB) deal with the safety measures to be adopted in railway tunnels in a special manner: " Sichereith in bestehenden Tunnels, Infrastrukturmassnahmen zur Erleichterung der Selbstrettung " and "Sicherheitsstandards für unterirdische Verkehrsanlagen der SBB” (the latter is now out of date). Recently, the Federal Transport Office (OFT) has instructed the railway companies to propose new measures to improve safety standards in long Swiss railway tunnels, since it has considered not appropriate to establish generalised standard measures for all tunnels.

1.3.2.1 Raccomandazione comune delle autorità di vigilanza sulle ferrovie della Germania, dell’Austria e della Svizzera in merito alla sicurezza dei viaggiatori in gallerie ferroviarie molto lunghe, 24.09.1992

Swiss Federal Office of Transport
Summary:
This recommendation refers to the safety of travellers in very long railway tunnels such as those designed for the Brenner, Gotthard and Loetschberg tunnels. First of all it aims at producing a harmonised approach to the concept of safety and presents a series of safety principles. The recommendation applies to new railway tunnels longer than 25 km.

1.3.2.2 Railway regulation: application rules, 01.01.1994

(Swiss Federal Railways).
Summary:
This document derives some safety principles for travellers in very long tunnels.

1.3.2.3 Weisung I-AM-EB-31/00: Sichereit in bestehenden tunnels, Infrastrukturmassnhamen zur Erleichterung der Selbstrettung, 06.12.2000

(Swiss Federal Railways).
Summary:
This document sets out the technical standards and functional requirements for the hardware which allows people directly involved in an incident to initiate an emergency response. The standard represents the application to the Swiss railway network of those safety criteria envisaged for authorized existing tunnels on the basis of the analysis of risk factors conducted by the Board of Directors of the SBB on 28/10/1998. The technical standards set out therein are valid for the design and improvement of the equipment and systems used for facilitating “self-help emergency action”, giving a sufficiently advantageous cost/quality ratio.

1.3.2.4 Ausfuhrungsbestimmungen zur Eisenbahnverordnung, 01.2001

Summary:
This regulation gives the reference law for railway infrastructures. For safety in Swiss railway tunnels, reference must be made to the various laws and, in particular, to the regulations given above, although it should be noted that the requisites set out therein are no longer regarded as sufficient by the Swiss authorities.

1.3.2.5 Verordnung vom 27 Februar 1991 über den Schutz vor Störfällen“ (Störfallverordnung, StFV, SR 814.012)

Summary:
This decree aims to protect population and environment from serious consequences following major incidents.

1.3.2.6 Verordnung vom 23 November 1983 über Bau und Betrieb der Eisenbahnen (Eisenbahnverordnung, EBV, SR 742.141.1)

Summary:
The present decree gives regulations on construction, operation and maintenance of the railway structures, systems and vehicles. It is mainly focused on railway safety.

1.3.3 Germany

Richtlinie
Anforderung des Brand- und Katastrophenschutzes an den Bau und Betrieb von Eisenbahntunneln (01.07.1997 with ammendments from 30.07.1999)

Eisenbahn-Bundesamt (EBA)
Summary:
This document was drawn up by the technical staff of the various Lander (regional authorities) and by a working party made up of executives from the Fire Brigade, DB SpA and the Federal Railway Office. The directive defines the safety measures necessary for the construction and operation of tunnels in order to guarantee (in railway tunnels) not only the actuation of emergency measures by travellers and personnel but also the use of rescue vehicles/equipment. This directive applies to new railway tunnels and does not cover urban line tunnels. For purposes of the document “tunnels” refers to engineering works whose length exceeds 500 m. In particular, the document covers new tunnels with a length (L) between 1 km and 15 km. Tunnels are classified by length: long tunnels (1 km ≤L≤ 15 m), very long tunnels (L>15 km). The minimum safety requirements increase with the length class. For tunnels in excess of 15 km measures must be evaluated on a case-by-case basis. In the case of existing lines, the appropriateness of applying such measures has to be evaluated in relation to the level of safety provided. However, the directive only applies when substantial elements of an existing tunnel are intended to be modified in the context of modernisation/retrofitting design work. The same principle applies to measures referring to service reorganisation undertaken to reduce operating costs.

1.3.3.1 Leitfaden für den Brandschutz in Personenverkehrsanlagen der Eisenbahnen des Bundes und der Magnetschnellbahn Jan. 2001

Eisenbahn-Bundesamt (EBA)
Summary:
This guidance document is concerned with fire protection measures in facilities with passenger traffic. These are defined as Station buildings, platforms and access ways to and from platforms. It is not specifically concerned with tunnels, but does contain general fire safety regulations which should also be followed in underground stations. It has not been included in the detailed comparison which follows, as this concentrates on running tunnels.

1.3.3.2 Anforderungen der DB Station&Service AG an den Brandschutz in Personenverkehrsanlagen - Draft Version 15.03.2001

Deutsche Bahn AG (DB)
Summary:
This draft document supplements the EBA Leitfaden above [1.3.3.2] and is also concerned with passenger facilities. Again, it has not been included in the detailed comparison, which concentrates on running tunnels.

1.3.4 France

1.3.4.1 Instruction technique interministérielle relative à la sécurité dans les tunnels ferroviaires (08.07.1998)

Ministère de l'Equipement, des Transportes et du Logement
Summary:
This directive mainly addresses questions of infrastructure, but it also considers rolling stock and operational procedures. The document deals with urban lines, mixed traffic lines and passenger traffic lines (modern or upgraded rolling stock) or freight lines during periods when there is no passenger traffic. The document allows for alternative safety solutions in addition to established minimum safety measures, but they must, in any case, be approved by competent authority. The diractive covers new tunnels whose length is between 400 m and 10 km. For tunnels of between 400 and 800 m there are lesser requirements. For tunnels in excess of 10 km (or 5 km for the "Railway Motorway") safety measures must be evaluated on a case-by-case basis.

1.3.4.2 Décret relatif a la sécurité' du réseau ferre national (30.05.2000)

Ministère de l'Equipement, des Transportes et du Logement
Summary:
This decree, referring to safety on the national railway network, is addressed to the French Railway Network (RFF), the National Company of French Railways (SNCF) and appointed railway companies. It sets out provisions that must be applied in the specification, design and construction of all systems comprising the infrastructure, technical and safety plant and rolling stock that belongs to or is used on the National Railway Network. The document is divided into three parts: the first concerns the construction of systems belonging to the Railway Network or for its use. The second refers to railway operating safety and the third sets out derogations from the decree's scope of application.

1.3.5 Norway

1.3.5.1 Substructure, regulations for new lines, tunnels – safety requirements, (01.01.2000)

Jernbaneverket
Summary:
These regulations cover all safety measures which must be considered when designing railway tunnels. The requirements are to be understood as the absolute minimum requirements. Whenever there are exceptional situations, such as tunnel stations or dangerously heavy goods traffic, then additional, case-specific measures should be adopted. These regulations only apply to newly-built tunnels with a length over 1 km. The document is divided into 6 chapters. The first three provide an introduction to the regulations, their scope and the classification of tunnels into 4 classes (A-B-C-D) by length, average daily traffic (trains/day) and maximum daily traffic (train/h). The fourth chapter lays down, with a specific table, the minimum safety requirements for each class of tunnel. The regulations also set out a table listing the supplementary measures to be adopted in various circumstances along with a description of the minimum requirements. The final two chapters provide indications for the maintenance of plant and equipment and for drawing up emergency plans.

1.3.6 Austria

General comments
In Austria there are regulations for the construction of tunnels called "HL Richtlinien" or "Guidelines for High Performance Lines". These regulations refer to the geometry of the basic infrastructure, the signalling systems and the basic technological systems of the tunnel. Furthermore, the authorities and the Fire Brigade have issued recommendations, which are also mandatory, that refers to infrastructure, technological systems and operation (“Guidelines for the construction and operation of railway tunnels”). The Austrian Federal Railways (OBB) also have regulations for training train personnel. In general the foregoing regulations and the recommendations are sufficient even if there is a need to integrate the various subjects together. The Austrian Federal Railways class tunnels by accident risk and importance, and for every class, specify specific risk factors (length, infrastructure, technological systems, traffic density, number of passengers).

1.3.6.1 Richtlinie

Bau und Betrieb von neuen Eisenbahntunneln bei Haupt- und Nebenbahnen, Anforderungen des Brand- und Katastrophenschutzes (1. Ausgabe 2000)

Österreichischer Bundesfeuerwehrverband
Summary:
This document forms the reference regulations for preparing opinions and conducting surveys as well as for decisions (by the competent authorities) on procedures for the construction and operation of railway tunnels. It applies to new tunnels whose length is between 1.5 km and 25 km. The document classes the tunnels as: short (L$\leq$ 1.5 km), medium (1.5 km $\leq$ L < 15 km), long (5 km $\leq$ L $\leq$ 25 km) and very long (L > 25 km). Very long

tunnels need safety measures that must be defined on a case-by-case basis. For existing tunnels, the degree to which each of the the provisions can be applied must be verified.

1.3.6.2 Eisbav: Eisenbahn-ArbeitnehmerInnershutzverordnung

Verkehrs-Arbeitsinspektorat
Summary:
The railway companies' guidelines already contain many regulations concerning the protection of employees working on the trackside. These safety measures have been standardized and summarized in a decree of safety for railway employees. In these instructions, the Transport Department for safety at work brought together some explanations of the safety decree for railway employees in order to make the practical application of the regulations easier. The first edition of these instructions was published in spring 2000. They are a simple and easily readable aid for employers and employees.

1.3.6.3 Allgemeines Sicherheitskonzept für mittlere Tunnel

OBB- Austrian Federal Ministry of Transport - Eisenbahn – Hochleistungsstrecken AG
Summary:
This document contains the criteria for the preparation of a general operational safety plan for existing medium-length tunnels, that is between 1.5 and 15 km. In particular, it defines methods for analysing risks and implementing possible measures for improving safety conditions. The document is divided into four chapters: risk evaluation, the examination of safety provisions, the evaluation of provisions, and conclusions.

1.3.6.4 Allgemeines Sicherheitskonzept für mittlere Bestandestunnel

OBB- Austrian Federal Minstry of Transport - Eisenbahn – Hochleistungsstrecken AG
Summary:
This document contains the general safety regulations for existing, medium-length tunnels (1.5 – 15 km). The contents of the document can be applied in a simplified form when planning for safety in newly-designed tunnels.

1.3.6.5 Allgemeines Sicherheitskonzept für lange Tunnel

OBB- Austrian Federal Ministry of Transport - Eisenbahn – Hochleistungsstrecken AG
Summary:
Not available.

1.3.6.6 Richtilinien für das Entwerfen von Hochleistungsstrecken

(HL- Richtilinien)
Summary:
These regulations deal with geometry, basic infrastructure and basic tunnel equipment.

1.3.6.7 Handbuch „Feuerwehreinsatz im Gleisbereich “Allgemeines Sicherheitskonzept für mittlere Bestandestunnel

ÖBB- ÖBFV
Summary:
Not available.

1.3.7 Spain

1.3.7.1 Instruccion obras subterraneas (01.12.1998)

RENFE - Ministirio de Fomento
Summary:
This document issued by the Spanish National Railway Network (RENFE) indicates objectives for improving safety and possible approaches to adopt in risk analysis and evaluation. The instructions, which mainly focus on prevention, describe a series of measures to be adopted, divided into standard and supplementary measures.
The instructions are divided into 5 chapters. The first presents the essential requirements for engineering works and general considerations on their classification by geometry and the construction method. The second concerns the obligations of the network operator during an emergency or maintenance. The third and fourth chapters stipulate the criteria to follow in the design and construction phases. The last chapter sets out the criteria for the installation of the various items of equipment and for the operational phase.

1.3.7.2 Medidas de seguridad en nuevos tuneles ferroviarios (04.2000)

RENFE
Summary:
These specifications describe the type and objectives of the minimum safety measures which must be adopted in order to facilitate the following of emergency procedures by passengers and railway personnel inside railway tunnels. The document also lays down the intervention procedures to be followed by non-railway rescue teams, without prejudice to any further safety measures. The subject matter covers measures to prevent risks and limit damage, as well as emergency response to be deployed by travellers/ on-board personnel, and how to exploit the intervention of external rescue facilities to the full. These measures are mainly directed towards the protection and evacuation of people. The scope of application for the specifications covers new urban railway tunnels but excludes underground railways. Their application to existing tunnels would be contrary to the principle of proportionality and for this reason application to existing tunnels is only justified when/if they are to be upgraded.

1.3.8 United Kingdom

1.3.8.1 «Railway Safety Principles and Guidance, Part 2, Section A, Guidance on the infrastructure». Chapter 5 : TUNNELS

UK Health and Safety Executive
Summary:
The document gives official guidance on safeguarding safety and health in railway environments. Chapter 5 provides the most direct reference to problems concerning safety in metropolitan tunnels and long tunnels (L≥1.500 m). The regulations indicate the factors to bear in mind in designing tunnels. While the final geometrical configuration of the tunnel will depend on the type of rolling stock to be used, the means foreseen for evacuation and the possibility that the tunnel is an extension of an existing tunnel. Usually a double bore configuration is envisaged for metropolitan and long railway tunnels. Solutions that derive from the internal division of a larger tunnel may be acceptable. The scope of application of the regulations is metropolitan line and railway tunnels whose length is L≥ 1.500m.

1.3.8.2 «System Safety Requirements for New and Re-opened Tunnels»

Railway Group Standard- Railway Safety GC/RT 5114 Draft 3f December 2002
Summary:
This document sets out the system safety requirements to be considered for new and re-opened tunnels.

1.3.8.3 «Guidance on System Safety Requirements in New and Re-opened Tunnels»

Railway Group Guidance Note- Railway Safety GC/GN 5614 Draft 2f December 2002
UK Health and Safety Executive
Summary:
This document provides guidance on the system safety required in tunnels to ensure their safe operation in support of GC/RT5114.

1.3.9 The Netherlands

1.3.9.1 The Dutch Vision on Safety in Road and Rail Tunnels (Draft)

Ministries of transports and of Inland Affairs
Summary:
The Dutch Tunnel Safety project group has conducted a detailed study on this question, published with the title “Tunnel ahead”, which included the cooperation of the Ministry of Transport and the Ministry of the Interior. The document mainly addresses the following problems: the lack of transparency in decision-taking procedures regarding options for providing safety in tunnels, the lack of standards and an adequate level of safety in tunnels, safety management during the operation of tunnels, and the organisation of safe procedures to be followed by engine drivers and on-board personnel in tunnels (training courses and periodic drills).

1.3.10 Sweden

1.3.10.1 Säkerhet i järnvägstunnlar, Ambitionsnivå och värderingsmetodik, Handbok BVH 585.30 (01.09.1997) - Zusammenfassung auf Englisch

Banverket
Summary:
The handbooks sets out to establish and identify a clear method for ascertaining that an adequate level of safety has been reached in tunnels and specifies the various subjects involved. The handbook refers to newly built tunnels but it can also be consulted for the purpose of verifying traffic safety in existing tunnels.

1.3.11 Finland

1.3.11.1 Technical Regulation and Guidelines for Railways” “Railway Tunnels”

RAMO, Finnish Rail Administration Board
Summary:
This document deals with all aspects of tunnel design, from general principles for tunnel design to the requirements for the rails, from the structural size of engineering plant to guidance on construction and maintenance. In the part covering design, the risk-analysis procedures to be followed are set out along with an analysis of the safety requirements in the event of an accident (material, plant, communication, power-supply requirements etc.)

1.3.12 USA

1.3.12.1 NFPA 130 Standards for Fixed Guideway Transit and Passenger Rail Systems

National Fire Association
Summary:
This standard covers fire protection requirements for passenger rail, underground, surface, and elevated fixed guideway transit systems including trainways, vehicles, fixed guideway transit stations, and vehicle maintenance and storage areas; and for life safety from fire in fixed guideway transit stations, trainways, vehicles, and outdoor vehicle maintenance and storage areas. Fixed guideway transit stations pertain to stations accommodating only passengers and employees of the fixed guideway transit and passenger rail systems and incidental occupancies in the stations. This standard establishes minimum requirements for each of the identified subsystems. The purpose of this standard is to establish minimum requirements that will provide a reasonable degree of safety from fire and its related hazards.

2 COMPREHENSIVE LIST OF SAFETY MEASURES

2.1 General design characteristics

G1 Range of Applicability
    G11 Limits on Applicablity
G2 Geometric configuration
    G21 Geometric configuration of new tunnels

2.2 Structural measures relevant to safety

S1 Emergency passenger exits
    S11 Parallel service and safety tunnels
    S12 Emergency cross-passages (distance)
    S13 Safe places
    S14 Escape routes
S15 Vertical exits/rescue shafts
S16 Lateral exits/access tunnels

S2 Emergency access for rescue staff
    S21 Tunnel access for emergency vehicles
    S22 Rescue forces' emergency vehicles (train, bimodal…)

S3 Drainage of flammable liquids
    S31 Inclination of tunnel axis
    S32 Track drainage system (drainage and retaining basins)

S4 Open areas
S41 Rescue areas

2.3 Safety equipment

E1 Smoke control ventilation
    E 11 Natural ventilation
E 12 Forced ventilation

E2 Emergency exit and rescue access ventilation
E 21 Emergency exit/rescue access ventilation

E3 Lighting systems
    E31 Emergency tunnel lighting
    E32 Emergency exit/rescue access lighting

E4 Escape signs in tunnels
    E41 Pedestrian exit signs
    E44 Other

E5 Communication and alarm system
    E51 Emergency telephones
    E52 Alarm push buttons
    E53 Fire/smoke detection

E54 Radio rebroadcast
E55 Loudspeakers

E6 Operation and traffic management
E61 Speed and density monitoring
E62 Traffic classification and control
E63 Tracking status of the train before entering a tunnel

E7 Power supply
E71 Power supply

E8 Fire suppression
E81 First and fire fighting
E82 Fire extinguishing systems (in technical rooms)
E83 Other

E9 General safety equipment
E91 General Safety Equipment (Running Tunnels)
E92 General Safety Equipment (Cross-passages)

2.4 Structure & equipment response to fire

R1 Reaction to fire
R11 Reaction to fire
R12 Fire protection requirements for structures
R13 Fire resistance requirements for equipment
R14 Additional measures

2.5 Emergency management

M1 Organisational measures
M1 Safety plans

3 MATRIX OF GUIDELINES CONTENTS

❖ Matrix of Guidelines Contents

Overview of Contents		Italy	Switzerland	Germany	France	U.K.	U.S.A.
	Available national guidelines:	**8**	**4**	**4**	**2**	**3**	**1**
Category	**Element *(safety measure)***						
General design characteristics							
G1 Range of Applicability	G11 Limits on Applicability	A,1,8	A,1,3	A,1	A,1	A,1,2,3	A,1
G2 Geometric configuration	G21 Geometric configuration of new tunnels	A,8	A,,3, !	A,,1,4	A,,1	A,1,2,3	A,1
Structural measures relevant to safety							
S1 Emergency passenger exit	S11 Parallel service and safety tunnels	A,1	0	A,1	A,1	A,2,3	0
	S12 Emergency cross-passages (distance)	A,1,8	!	A,1,4	A,1	A,1,2,3	A,1
	S13 Safe places	A,1	A,1	A,1,4	0	A,2,3	A,1
	S14 Escape routes (lateral walkways)	A,1	A,3,!	A,4	A,1	A,1	A,1
	S15 Vertical exits/rescue shafts	0	0	A,1,4	0	A,1	A,1
	S16 Lateral exits/access tunnels	A,1	0	A,4	A,1	A,1,2,3	A,1
S2 Emergency access for rescue staff	S21 Tunnel access for emergency vehicles	A,1	A,3	A,1,4	A,1	A,1	A,1
	S22 Rescue forces emergency vehicles (train, bimodal..)	A,1	!, A,1,2	A,1,4	A,1	0	0
S3 Drainage of flammable liquids	S31 Inclination of tunnel axis	0	0	A,1,4	0	0	0
	S32 Track drainage system (drainage and retaining basins)	0	A,3	0	A,1	A,1	0
S4 Open areas	S41 Rescue areas	A,1	A,3	A,1,4	A,1	A,1	0
Safety equipment							
E1 Smoke control ventilation	E11 Natural ventilation	A,1	0	0	A,1	0,A,1,2,3	-
	E12 Forced ventilation	0, A,1	!	0	A,1	0,A,1,2,3	A,1
E2 Emergency exit and rescue access ventilation	E21 Emergency exit / rescue access ventilation	A,1	0	A,1,4	0	0	0
E3 Lighting systems	E31 Emergency tunnel lighting	A,1	A,3	A,1,4	A,1,!	A,1	A,1
	E32 Emergency exit / rescue access lighting	A,1	0	A,4	A,1	A,1	A,1
E4 Escape signs in tunnels	E41 Pedestrian exit signs	A,1	A,3	A,4	A,1	A,1,2,3	A,1
	E42 Other	A,1	0	0	0	0	0
E5 Communication and alarm system	E51 Emergency telephones	A,1	A,3	A,4	A,1	A,1,2	A,1
	E52 Alarm push buttons	0, !	0	A,4	0	0	0
	E53 Fire/smoke detection	0, !	0,!	0	0	A,2,3	A,1
	E54 Radio rebroadcast	A,1	A,3	A,1,4	A,1	A,1,2,3	A,1
	E55 Loudspeakers	A,1	0	0	0	0	0
E6 Operation and traffic management	E61 Speed monitoring and intensity	0	!	A,1,4	A,1	0	0
	E62 Traffic classification and control	O, !	!	A,1,4	A,1	0	0
	E63 Tracking status of the train before entering a tunnel	0	0	0	0	0	0
E7 Power supply	E71 Power supply	A,1	A,3	A,4	A,1	A,1	A,1
E8 Fire suppression	E81 First and fire fighting	A,1	A	A,1,4	A,1	A,1	A,1
	E82 Fire extinguishing systems (in technical rooms)	0	A*	0	0	0	0
	E83 Other	0	0	0	0	0	0
E9 General safety equipment	E91 General Safety Equipment (Running Tunnels)	A*	A*	A*	A*	A*	-
	E92 General Safety Equipment (cross-passages etc)	A*	A*	A*	A*	A*	-
Structure & equipment response to fire							
R1 Reaction and resistance to fire	R11 Reaction to fire	A,1	!	A,1,4	A,1	A,1,2,3	A,1
	R12 Fire protection requirements for structures	A,1	A,3	A,1,4	A,1	A,2,3	A,1
	R13 Fire resistance requirements for equipment	A,1	A,3	A,1,4	A,1	0,A,2,3	A,1
	R14 Additional measures	0	0	0	0	0	0
Emergency Management							
M1Organizational measure	M11 Safety plans	A,1,7	!, A,3	A,1,4	0	0	A,1

Legend:	0: No requirements, little information. ! : Information derived directly from the national railway network.	A: Normative/guideline information N: national document number (ref. Chapter 1)	*: Information derived from the whole of available norms / regulations.

Figure 1 - Overview (matrix) giving types of requirements, amount of information and reference to the documents.

Notes:

The table shows only those parameters which appear to be dealt with consistently in the various national documents.

The numbers used are a reference to the individual National guidelines, see Chapter 1 §1.1. The same convention is used to refer to the national regulations/guidelines in the column headed “ref.” of each of the *element tables* given in chapter 4.

4 DETAILED COMPARISON

The detailed comparison comprises a description of the role of each safety measure, comparison of the requirements in form of direct quotes from the text of the guideline *([…])* or sometimes in form of a synthesis of the different available documents *([*])*, or as information directly derived from the National Railway network *([!])*.

The matrix shown in Figure 3.1 gives a comprehensive list of 39 safety measures in five categories: G: General design characteristics, S: Structural measures relevant to safety, E: Safety equipment, R: Structure & equipment response to fire, and M: Emergency management. The main categories have been subdivided into 2, 11, 21, 4 and 1 categories respectively. For each of the elements identified, the reference is made to the national guidelines of 6 selected countries (5 western European countries + USA). Where available, information from some other European countries including Spain, Finland, Sweden, Norway and Denmark is also provided. This has been obtained from the National Railway Infrastructure Manager. This additional information is given in rows titled *"other + name of the country".*
Each measure for detailed comparison is prefaced by general comments that define the role of the measure, and that mostly reflects the point of view of the International Union of Railways (UIC). All the comparisons are followed by quotations from UN/ECE Recommendations.

4.1 General Design Characteristics

4.1.1 G1 – Range of Applicability

4.1.1.1 Role of the measure

As a rule, regulations set a minimum length for tunnels to which they apply. The necessity of tunnel-specific measures is often indicated for very long tunnels. The regulations are generally applied to newly constructed tunnels, and may be applicable (either fully or in part) when relevant restructuring works are required on existing tunnels or those under construction.

4.1.1.2 Synthesis – comments

Analysis of the various regulations reveals that the range of applicability of safety regulations is similar for almost all of the principal western European countries. Italy, France, Germany, Austria and the U.K. have similar ranges over which their regulations apply, beginning between 1 km to 1.5 km and extending to 20 km to 25 km for new tunnels.
Among the various regulations on new construction of tunnels, those of Switzerland give only general guidelines on the subject of safety, these guidelines being applied to particularly long tunnels (L>25 km). For existing tunnels, only the Italian regulations precisely specify the range of applicability for the standards contained in their guidelines, giving a detailed illustration.
In this regard, other European regulations are limited simply to suggesting the possible application of the 'new-build' standards, but only in case of substantial modifications. Unlike Germany, Austria, France and the U.K., Switzerland provides a special regulation covering all existing tunnels, which gives infrastructure standards to facilitate rescue services.

4.1.1.3 Comparison tables

- G11 Limits on Applicability

Country	Ref.	Requirement	Comment
Italy	[1,8]	Tunnels on mixed traffic lines. Existing tunnels and those in course of construction in 1997, having lengths of between 5 and 20 km. Applicability extended (according to the RFI note of 17/12/2001) to future new tunnel construction with lengths between 2 and 20 Km. Long base tunnels (L>20 Km) must have individual study. The document does not take into consideration stations in tunnels or Metro tunnels, for which there is a specific regulation.	
Switzerland	[1,3]	[1] New tunnels having lengths of over 25 km New and existing tunnels (Weisung I-AM-EB 31/00) – 06.12.2000	
Germany	**[1]**	New tunnels having lengths of between 1 and 15 km For tunnels of over 15 km, individually tailored measures shall be provided. Possible application to existing tunnels, but only in the case of substantial modifications	
France	[1]	Tunnels for urban routes, passengers and mixed New tunnels, having lengths of between 400 and 10,000 m A few measures are required for tunnels between 400 and 800 m Safety measures must be evaluated on a case-by-case basis for tunnel lengths over 10 km. or lengths over 5 km. for heavily used railway lines.	
U.K.	[1,2, 3]	[1] Low-cover (shallow) railway tunnels. [1] Tunnels for metro lines and heavy rail, having lengths of ≥1.5 km. [2,3] New tunnels and re-opened tunnels	
U.S.A.	[1]	(1-1.1) …This standard establishes minimum requirements for …passenger rail, underground, surface, and elevated, fixed guidepath transit systems including tramways,…	
UNECE	[10]	These measures shall be applied to all railway tunnels. They may be reduced for short tunnels (<1 km), but shall be adapted or increased for very long tunnels (>15 km), undersea tunnels and steep mountain tunnels. Measures for existing tunnels, requiring alterations through civil works, may only be applied at reasonable costs during significant reconstruction works. However, safety requirements must also be achieved through measures regarding rolling stock and management.	
Other Austria		New tunnels, having lengths between 1.5 and 25 km Very long tunnels (L>25 km) require safety measures to be defined case by case. The extent the provisions provided can be applied to existing tunnels has yet to be verified.	
Other Norway		Tunnels, having lengths of over 1km 3 categories of tunnels (A,B,C) shall be identified based on the length, trains/day and maximum timetable frequency.	

4.1.2 G2 – Geometric configuration

4.1.2.1 Role of the measure

For new projects, there is a clear trend among authorities to define minimum escape distances and requirements / precautions for mixed traffic. If these requirements cannot be fulfilled with double-track tubes and escape facilities, double-tube single-track tunnels may be the solution. In any case the definition of the tunnel system - double-track tube or double-tube, single-track tunnel - is often a multicriteria decision based on: construction costs, construction time and risks, operation (maintenance concept, crossovers), topography (including space at the portals), aerodynamic aspects and safety.

4.1.2.2 Synthesis – comments

Analysis was performed with reference to mixed-traffic lanes (the righthand column in the table). A comparison reveals that Italy, Germany, France and the U.K. have regulations that clearly indicate the geometric configuration that must be adopted for new tunnels, based on traffic considerations and on the length of the tunnel to be constructed. Swiss and Austrian regulations do not address the subject explicitly.
The configuration suggested in some regulations is double-tube, starting from lengths below 1 km in Italy, this configuration is only 'recommended' for tunnels of 1 km, but is compulsory for tunnels of 2 km or more. Emphasis is placed on the fact that, within the context of safety, both the single-tube double-track and double-tube single-track solutions have advantages and disadvantages. The double-tube solution may be safer, since it prevents the risk of derailment accidents, which obstruct the adjacent track, and allows use of the track, not involved in the accident, as a safe haven. On the other hand, the single-tube configuration supplies more space for rescue operations, and also a greater volume into which fire and smoke can spread (giving more time for evacuation). For high-speed trains, the single-tube double-track solution could be preferable, whereas for mixed traffic, the double-tube configuration may be more suitable, keeping aerodynamic factors in mind (UN/ECE Recommendation). In any case, international organisations (UN/ECE and UIC) find that the choice should be the result of a thorough evaluation of all parameters related to safety as well as accurate economic assessments.

4.1.2.3 Comparison tables

- G21 Geometric configuration of new tunnels

Country	Ref.	Requirement	Comment
Italy	[1,8]	Single tube or double-tube, without distinction, down to 1,000 m Single tube or double-tube based project specificity, focussing on double-tube solutions for lengths of between 1,000 and 2,000 m Double-tube for tunnel lengths of over 2,000 m	Parameters comprised within category G2 have been compared with reference to **mixed traffic operations.** Double track – single tube: L < 2.000 m Single track – single tube: L ≥ 2.000 m
Switzerland	[3,!]	The choice of the single-tube or double-tube tunnel system should be the result of a thorough evaluation that takes into consideration all parameters related to: safety, the level of traffic and length of the line as well as cost/benefit economic considerations; thus, a field of lengths for differentiating the most suitable solution is not defined a priori [1]	[[!] Information obtained directly from railway network.
Germany	**[1,4]**	[4] On double-track lines, long and very long tunnels are to be designed as parallel, single-track tubes, when the operating programme provides for unrestricted mixed operations of passenger and freight trains. [1] The double-tube configuration for mixed-traffic lines must be adopted for tunnel lengths of over 1,000 m.	Double track – single tube: Undefined Single track – single tube: L ≥ 1.000 m
France	[1]	The adoption of the single-tube double-track configuration or double-tube single-track configuration depends on the type of traffic and length of the tunnel. The double-tube configuration, starting at lengths of 800 m., is adopted for mixed-traffic lines.	Double track – single tube: Undefined Single track – single tube: 800<L<10.000m for mixed-traffic lines, ad hoc solutions for L ≥ 10.000m
U.K.	[1,2, 3]	[2]….The choice between two single track tunnels or one double track tunnel shall be justified by a risk assessment, so as to provide acceptable safety during normal, degraded and emergency operating conditions. [1] As a rule, double-tube configuration is foreseen for long railway tunnels (>1,500 m). [1] Double-tube solutions, derived from an internal division of a broader tunnel, are acceptable (divider partition).	
U.S.A.	[1]	Various types of section are taken into consideration: double bore with connections, single bore with a partition and simple single tube.	

❖ Detailed comparison

UNECE	[10]	(C.1 01) …both single-tube double-track and double-tube single-track tunnels have their advantages and disadvantages. Double-bore single-track tunnels might be safer as they avoid accidents caused by derailments obstructing the adjacent track and they provide the second tube as a possible safe haven. …..double-track tunnels have more space for possible rescue operations but they also have more space for smoke and fire to spread. …. The choice should be the result of a thorough evaluation of all parameters (such as, for example, length of the tunnel, type of traffic, etc.) related to safety as well as cost considerations.	

4.2 Structural measures relevant to safety

4.2.1 S1 - Emergency passenger exits

4.2.1.1 Role of the measure

Emergency exits for tunnel users are installed with the purpose of having a safe haven in case of accidents in the tunnel. The exits will mainly be used in case of a fire in the tunnel. The emergency exit can be connected to an adjacent running tube, to a dedicated service tunnel or may lead to the outside. The connection can be direct or through a cross passage, shaft or similar. In some cases shelters are arranged as safe havens, where tunnel users can stay temporarily.

4.2.1.2 Synthesis – comments

All of the guidelines indicate that the adoption of a parallel tube as a smoke-free escape route is generally required for new tunnels, with the distance between emergency exits varying from 250m to a maximum of 500m. For new, shorter tunnels, a double-tube single-track configuration is generally preferred and the distance between emergency exits in this latter case may vary from 1000m to 2000m.
For newly constructed, single-tube, double-track tunnels, exits (vertical and/or horizontal: shafts, service tunnels, etc) are generally provided, which lead directly outside, with maximum separation distances varying from 500 to 1,000m. In practice, the distance actually adopted between exit passages varies depending on the local situation, operating conditions and the view of the safety concept based on the applicable regulations.

4.2.1.3 Comparison tables

- S11 Parallel service and safety tunnel

Country	Ref.	Requirement	Comment
Italy	**[1]**	Mentioned but not prescribed. (3.1.3) If a single-tube, double-track solution is chosen for new tunnels having lengths of less than 2,000m, a parallel service tunnel shall be provided that is equipped with appropriate connections or, alternatively, protected passageways shall be built that lead directly to the open. A double-tube solution with cross-passages every 250m shall be chosen for tunnels having lengths of over 2,000m.	
Switzerland		No reference.	
Germany	**[1]**	Mentioned but not prescribed. The whole concept is defined by the main rule that a safe place must be accessible within 500m of each point in the tunnel. 'Safe places' are defined as portals, escape tubes, escape shafts and cross passages or passageways that lead to an escape tube, escape shaft or the other tube. If the tunnel is shorter than 1km (category 1) then the above- mentioned measures are not mandatory. If it is longer, a combination of the above mentioned measures (which are then called emergency exits) is needed. If an escape tube is used, then it must have a minimum cross section of 2.25 x 2.25m, a longitudinal gradient < 10%, and a length of no more than 150m if it ends in an escape shaft If the length is more than 300m, it must allow vehicular access.	
France	[1]	(4.1.3)…Where parallel escape tube is available, it must be provided with cross passages at least 2.2m tall, 4 'passage units' wide and no more than 800 m apart. The cross passages must have 2 doors (fire resistance of 120 minutes, minimum width of 2 'passage units')	
U.K.	**[1,2,3]**	[1] Mentioned but not prescribed. [2,3] For long tunnels the place of relative safety could be a parallel tunnel, a service tunnel, specially constructed underground spaces, cross passages, etc. For shorter tunnels, an external place of relative safety, such as adjacent to the portal, would normally suffice.	
U.S.A.		No reference.	
UNECE	**[10]**	(C.3 10)…construction of a parallel service and safety tunnel… should be based on an assessment of the geotechnical and operating conditions and cost-benefit considerations for each tunnel. Possible benefits: pilot tunnel…, advance knowledge of …the ground for the main tunnel, logistic opportunities in construction and service, cable and pipe runs clear of the railway, maintenance access to technical rooms at any time.	

- S12 Emergency cross-passages (distance)

Country	Ref.	Requirement	Comment
Italy	[1,8]	For tunnels equipped with a parallel service tunnel for evacuating persons and for double-tube tunnels, connections shall be provided that guarantee sealing against smoke and flames, so as to make escape routes safe. (3.1.3.b). *Double-tube tunnels*: distance between cross-connections, 250m; *Single-tube tunnels*: maximum acceptable walking distance along a lateral walkway, 2,000m. Alternative solutions must be shown to guarantee equivalent safety criteria. (3.1.3.1.a). The following equipment and/or systems must be provided within the emergency cross-passages: • overpressure systems • telephones • illuminated signs • emergency lights • fire and smoke-proof doors fitted with anti-panic handles (crash bars).	
Switzerland	[!]	Cross-passages are provided in double-tube tunnels. Maximum distance between passages: 500m The following installations and/or systems are provided: • Emergency lights • handrails • signs • telephones • plenum systems.	[!] Information obtained directly from railway network.
Germany	**[1,4]**	For long and very long tunnels intended for mixed traffic, a double-tube system must be adopted. In this case cross-connections and the parallel tube can be used for escaping persons and rescue teams. A safe place must be available at a maximum distance of 500m from any point in the tunnel. Maximum distance between cross-connections: 1000m. These must be equipped with fire and smoke-proof, doors (fitted with anti-panic handles) that close automatically and are at least 1.0m wide. They must also be equipped with emergency lighting.	
France	[1]	(4.1.3) Where a parallel escape tube is available, cross passages must be provided (at least 2.2m in height, 4 'passage units' wide, maximum distance of 800m). The cross-passages must have 2 doors (fire resistance of 120 mins, minimum width of 2 'passage units') to keep both the cross passages and the adjacent tube free from smoke.	
U.K.	[1,2,3]	[1] (52).....Cross-passages between single-track running tunnels or to a service tunnel should be provided on the basis of safety assessments. Their max spacing is based on: • the length of train, • the method of evacuation; • the needs of the emergency services. …If …between the running tunnels, consideration should be given to: • the passage of smoke and heat; • the opening and closing of doors if provided; • risk to people from trains in any parallel tunnel, including any aerodynamic effects. The most common installations are: • Ventilation systems providing fresh air to the space and keeping it smoke-free • Fire and smoke-proof doors. [2,3] Distances between any cross passages ….should be determined by a risk assessment.	

U.S.A.	[1]	(3-2.4.3)…cross passageways shall not be farther than 800 ft (244 m) apart. Openings in open passageways shall be protected with fire door assemblies having a fire protection rating of 1½ hour with a self-closing fire door.	
UNECE	[10]	(C.3 09)…Cross passages should … connect the main tunnel with safe places. … constructed between the tubes of double-tube single-track tunnels or a double-track tunnel and a safety tunnel. … should be lit and have means of communication and be designed to prevent spreading of smoke into safe areas. … doors on exits to cross passages should be able to resist fire for 30 minutes and be able to resist the aerodynamic pressures found in the tunnel. ….. they should be easy to operate by hand or if heavy be motorized. In some cases where natural airflow does not exist, installing two doors (several metres apart) would ensure increased safety both by raising resistance to fire and by ensuring a pressurized environment.	

- S13 Safe places

Country	Ref.	Requirement	Comment
Italy	[1]	*Single-tube tunnels:* safe places must be able to be reached by means of emergency routes not longer than 2000m. *Double-tube tunnels:* the tube not involved in the accident constitutes a safe place with regard to the other tube and access shall be through cross-passages placed every 250m.	
Switzerland	[1]	[1] Protected areas are foreseen: Emergency exits towards the outside; tunnels or parallel service tunnels; underground emergency areas equipped with a fresh air supply and communication with the train control centre.	
Germany	**[1,4]**	Reaching a safe place is the objective of self-rescue. The portals (the open air) and the emergency exits after reaching the lock are deemed safe places. Safe places are reached via escape routes and are connected with each other by means of these. As an absolute principle, it must be possible to reach safe places after 500 m at the furthest from any place in the tunnel.	
France		Not mentioned	
U.K.	**[2,3]**	[2,3] (C.3) The capacity…should be determined by an assessment of potential incident circumstances, …. As an indication, …. it should be assumed that the capacity should be sufficient for the maximum potential occupants of a train at a space rate of at least 0.4 m^2 per person and an air supply rate of 30 m^3/hour per person.	
U.S.A.	**[1]**	Points of safety. See S15, S16. More detailed requirements requested for subways.	
UNECE	**[10]**	(C.3 06) It is recommended that a maximum distance between two safe places (portal of the tunnel, cross passage leading to another tunnel, emergency exit) be defined to enable easy and quick self-rescue. The exact distance varies depending on the local situation, operating parameters and the safety concept. In double-bore single-track tunnels and parallel safety tunnels, this distance should not exceed 500 metres. It is recommended to use cross passages between two parallel tubes rather than exits to the surface. Construction shafts and places close to the surface should be used for emergency exits.	

❖ Detailed comparison

- S14 Escape routes (lateral walkways)

Country	Ref.	Requirement	Comment
Italy	**[1]**	*Existing tunnels:* • (1.2.3.a) The maintenance passages should be used as lateral walkways: their width is about 0.5m on average. *Tunnels in progress*: • (2.2.3.1.a/2.2.3.2.a) The maintenance passages should be used as lateral walkways: their width is about 50 cm on average. • Also lateral tunnels used during excavation period, if present, will be used (6m wide and 5m high on av.). Such lateral tunnels should be divided into 2 sections (for vehicles and for pedestrians). The pedestrian walkway should be 1.2m wide, with lights at 2m above the floor that provide 5 lux at 0.1m from floor. *New tunnels*: • Lateral walkways: Walkway on one side of single - track tunnels and on both sides of double – track tunnels. Min. width 0.85m, optimal width 1.20m (new tunnels). Surface should be made of cast-in-place concrete or pre-cast slabs. Non-slip hard and even surface. Height 0.2m above the rail bottom. No handrail is required. The centreline of any cable duct should be 0.7 m from the tunnel wall or from alcove threshold. • (3.3.4.1.a/3.3.4.2) Also any lateral tunnels used during the excavation period, if present, will be used. Such tunnels will be divided into 2 sections (for vehicles and for pedestrians, see tunnel in progress).	The guideline includes: chapter 1 for 'existing tunnels', Chapter 2 for 'tunnels in progress', Chapter 3 for 'new tunnels to be realized'.
Switzerland	[3,!]	Lateral walkway must be free of obstacles, with adequate lighting, must be always be on the same side through the tunnel and, if possible, on the outer side of a curve. Handrail dimensions: Cross section: 120x40 mm; Height above escape route: 1.30 m; Distance between anchorages: 2.00 m. In narrow sections, the handrail may be replaced by touch-guide panels, the size of each section being 200x30 mm. The handrail must skirt any existing obstacles.	[!] Information obtained directly from railway network.
Germany	[4]	An escape route must be provided alongside obstacles and have adequate lighting. Escape routes must have clear headroom of at least 2.20 m. Escape routes must be at least 1.20 m wide. The width is the space between the widest rail vehicle standing with its doors open and the guiding system (handrail) located on the tunnel wall In suburban railway tunnels, it is permissible for the escape route to be limited to a width of 80 cm in the area of the stationary vehicle. Away from the vehicle, however, the width must be 1.20 m. Handrail must be present.	
France	[1]	(3.1.2) …In single track tunnels there must be a walkway on one side. For double track tunnels, there must be walkways on both sides, at a minimum of 0.1m above the top of the rail. Width must be 0.5m at the base and 0.7m at shoulder level and free of obstacles to a height of two meters. No handrail is required.	
U.K.	**[1]**	- (53.a)…. a side walkway to permit evacuation through ….side doors of the train should be provided. The side walkway should take into account the floor height and stepping distance from all types of train using the tunnel. - The walkway may be designed to provide derailment containment on that side. - (53.c) …. free of obstruction, at least 850 mm wide with 2000 mm headroom above the centreline of the walkway, even and anti-slip surface. Any change in level should be achieved by ramps with a gradient not steeper than 1 in 12. A continuous handrail should be provided between access points	
U.S.A.	**[1]**	(3-2.1.3) Walk surfaces designated for evacuation of passengers shall be constructed of non-combustible materials. Walk surfaces shall have a slip-resistant design.	

Other Spain		Min. width > 1.2 m (0.9 m at some points). In double-track tunnels walkways are required on both sides of the tunnel. Handrail(s) must be present.	
Other Finland		An uncluttered exit route should be arranged on both sides of the rail tunnel. The free width of these exit routes should be at least 1,600 mm and the free height at least 2,200 mm. The walk surface of the exit route should be on the rail level and sufficiently smooth. The exit route should be provided with a handrail as specified in Section 18.383.	
Other Sweden		Basic standard: Walkways on both sides of the tunnel. Height 0.6 m above rail top. Width 1.2 m. Hard and even floor, free of obstacles. No obstacles permitted within the walkway. Additional standard: Light coloured walls up to 2 m.	In Sweden the code on safety is risk based. Therefore all measures are divided in a basic standard and additional standard.
Other Norway		In all tunnels, the distance from the tunnel wall to a train shall be min. 1.5 m and with 2.2 m free height in this area. A walkway shall be placed on the same side as a handrail. If the ballast is to be used as the walkway, then the ballast level shall be the same as the height of the sleepers. Any cable duct shall be 0.7 m from the tunnel wall if the duct top forms part of the walkway.	
Other Denmark		In single-track tunnels: Walkways on both sides of the tunnel. In double-track tunnels: Walkways on both sides of the tunnel. Height 0.55 m above rail top (in double-track tunnels 0.35 m). Width 1.45 m. Obstacles in the walkway must be covered. No handrails are required.	
UNECE	**[10]**	Walkways on both sides of the double-track tunnels. Height depends on the specific tunnel situation. Width at least 70 cm and preferably 120 cm. Handrails at an appropriate height above walkway	

- S15 Vertical exits / rescue shafts

Country	Ref.	Requirement	Comment
Italy		No reference.	
Switzerland		No reference.	
Germany	**[1,4]**	[4] Rescue shafts are vertical structures with stairs which permit exit from the tunnel to the outside or access to the tunnel from the outside, and which generally lead away from/to an emergency exit. They are connected to the main tunnel via a lock and, possibly, a rescue passage. On the countryside, the rescue shaft ends in a shaft building in which also the fire and disaster protection equipment, such as the firewater supply, is located. The maximum height of rescue shafts must not exceed 60 m. Where the height of the rescue shaft is more than 30 m, a lift with the minimum cage dimensions of 1.1 x 2.1m must be provided in addition to stairs.	
France		No reference.	
U.K.	**[1]**	- [1] Emergency access points to a tunnel should be provided at distances determined by the ability of the fire brigade to penetrate effectively into the fire zone (see S14). - Current practice indicates that distances between access points should be in the order of 1 km where there are twin-bore tunnels with adequate intermediate cross-passages. In other circumstances this distance may need to be reduced	
U.S.A.	**[1]**	Emergency exit stairways shall be provided throughout the tunnels and spaced so that the distance to an emergency exit shall not be greater than 1250 ft (381 m) unless otherwise approved by the authority having jurisdiction.	
UNECE	**[10]**	(C3.07) Vertical exits/access should be provided in single-bore tunnels. They may be feasible only if the tunnel lies near the surface. It is recommended that vertical exits are equipped with proper lighting and communication means. Where the shaft is higher than 30 m, a fire-fighting lift might be installed. Stairways and lifts should be pressurized and/or equipped to ensure a smoke-free environment.	

- S16 Lateral exits / access tunnels

Country	Ref.	Requirement	Comment
Italy	**[1]**	See S11, S12, S14	
Switzerland		No reference.	
Germany	**[4]**	Lateral exits (rescue passages) may connect both the main tunnel with a rescue shaft, lead directly from the main tunnel to the open air and also be provided parallel to the main tunnel over its entire length. Any rescue passage longer than 300 m leading directly to the open air must be suitable for road vehicles. The cross-section of rescue passages must be at least 2.25 x 2.25 m. They must not be more than 150 m long if they do not lead directly but via rescue shafts to the open air. Rescue passages longer than 300 m must be suitable for motor vehicles. The gradient should not exceed 10%. A combination of rescue passages and rescue shafts is permitted.	
France	**[1]**	(3.1.1) ...access passages must allow for rescue vehicles to pass and cross the track, if necessary.	
U.K.	**[1,2,3]**	[1] Emergency access points to a tunnel should be provided at distances determined by the ability of the fire brigade to penetrate effectively into the fire zone. Current practice indicates that distances between access points should be in the order of 1 km where there are twin-bore tunnels with adequate intermediate cross-passages. [2,3] Doors should be provided to prevent smoke from spreading.	
U.S.A.	**[1]**	(3.2.4.1)…Emergency exits shall be provided from tunnels to a point of safety. See S15.	
UNECE	**[10]**	(C3.08) Lateral exits/access should be provided in single-bore tunnels, they should be located in the areas near the surface to limit their length as well as in places for easy exit and access by emergency services. The cross section of these exits should be determined on the basis of other safety elements but ideally their dimension should be 2.25 m x 2.25 m with a maximum length of about 150 metres. Lateral exits longer than 150 metres should be made accessible by road vehicles. The same installations that ensure a smoke-free, visible and otherwise safe environment in vertical exits should also be installed in lateral exits.	

4.2.2 S2 - Emergency access for rescue staff

4.2.2.1 Role of the measure

In case of an emergency, e.g. a fire or serious accident, rescue staff may not be able to access the accident site directly but may have to gain access through an adjacent tunnel (service or traffic tunnel) or through shafts. Access from the adjacent tunnel may make it possible to drive the vehicles (bimodals or trains) directly to the accident site. Rescuers may also enter on foot.

4.2.2.2 Synthesis – comments

The relevant fire brigades should have road/rail vehicles, which are able to run on tracks to convey staff and equipment rapidly to the accident site. The main goals are to support self-rescue, provide first aid and initial fire-fighting. Some countries use trainway vehicles to reach the site of the accident. Only in some countries are the operations coordinated by fire brigades. The tables below give descriptions of rescue vehicles and some features of safe places mandated by the different regulations/guidelines.

4.2.2.3 Comparison tables

- S21 Tunnel access for emergency vehicles

Country	Ref.	Requirement	Comment
Italy	**[1]**	*Existing tunnels* – *Tunnels already under construction:* (1.1.1) (2.1.1)..• *Access gate*: the area adjacent to the railway infrastructure is normally fenced off and, hence, inaccessible. An appropriate opening is required for rescue teams …. the so-called "access gate" should be linked to public roads and identified by a sign stating: "emergency access": This gate shall be generally kept closed ….. The gate shall be not less than 4m wide. • *Access road*: this road shall be not less than 4m wide and expand to 6 m every 250m to allow rescue vehicles to pass each other. Slope must be less than 16%. Bend radius of ≥ 11 m. *New tunnels:* As above, except: • *Access gate*: width not less than 6m. • *Access road*: width not less than 6m to allow for the movement of rescue vehicles in both directions. (3.2.2).	The guideline includes: Chapter 1 for 'existing tunnels', Chapter 2 for 'tunnels already under construction', Chapter 3 for 'new tunnels'.
Switzerland	[3]	Access to both ends of the tunnel, with rescue trains driven by railway staff. Areas for sheltering the injured and parking rescue vehicles are located near tunnel portals.	
Germany	**[1,4]**	See S16 If a lateral exit leads to a rescue passage longer than 300 m, which leads, in turn, directly to the open air, then it must be suitable for use by road vehicles. In case of two parallel single-track tubes, the safety concept will be based on use of the unaffected tube. In this case, escape and rescue measures can be carried out via the unaffected tube. Since it is not necessary to build rescue shafts in this case, the only means of access into the tunnel is via the portals. In order to minimise the length of travel for fire brigades to reach the fire, it must be possible for their vehicles to reach all cross-passages. Access routes to the tunnel portals must lead from rescue areas, when these are required. The access routes to the tracks must be secured by means of lockable barriers. This lock must be a fire brigade lock, since only the fire brigade is authorised to enter the tunnel with road vehicles.	
France	[1]	(3.1.1) …For 2-bore tunnels, facilities must be provided to let emergency vehicles pass from one bore to the other.	
U.K.	[1]	Rescue teams shall gain access from tunnel entrances and intermediary access ways, protected by a filter zone.	
U.S.A.	[1]	Rescue teams shall gain access from tunnel entrances and intermediary access ways.	

- S22 Rescue forces' emergency vehicles (train, bimodal…)

Country	Ref.	Requirement	Comment
Italy	**[1]**	- Bimodal road /rail vehicle from the Fire Brigade + roads and loading areas at each entrance - Railway carriages (type: MM 380 G) for rescue operations	
Switzerland	[!,1,2]	Trains equipped for extinguishing fires and for rescue operations, stationed at both ends of the tunnel.	[!] Information obtained directly from railway network.
Germany	[1,4]	- Normal vehicles from the Fire Brigade., bimodal road / rail vehicle + access ways, rescue areas - Rail wagons for conveying the injured out of or heavy equipment into the tunnel	
France	[1]	- (3.3.3)…At the tunnel entrances, two rail wagons equipped with a braking system must be available for rescue operations. They must have a minimum carrying capacity of 500 kg and allow for the evacuation of an injured party lying on a standard-size stretcher.	
U.K.		No reference.	
U.S.A.		No reference.	
UNECE	[10]	(C.4 12) …Fire brigade and other rescue services should get into the tunnel with their equipment as fast as possible, regardless of the type of vehicles used. In some places specialized rail vehicles are recommended as a part of the rescue concept. Rail/road vehicles for rescue are only recommended as a part of the comprehensive rescue equipment provided by the fire brigade. It is recommended that rescue trains be manned by the railway operator's staff and not fire brigade staff, who may not be familiar with the use of railway vehicles, equipment and special railway procedures. It is recommended that the fire brigade utilize either the road vehicles they use in their daily work or rail/road vehicles.	
Other Austria	[*][!]	Standard rescue vehicles that can only access the tunnel if: • a suitable traffic lane is available; • the vehicles allow for mixed road/rail use. • suitable vehicles from the railways are used	

4.2.3 S3 – Drainage of flammable liquids

4.2.3.1 Role of the measure

If flammable liquids are spilled in a tunnel, there is a risk that the spill will ignite and give rise to a serious fire. If the tunnel is well drained and the flammable liquids are collected in a system suitable for the purpose, this risk can be reduced.

4.2.3.2 Synthesis – comments

Drainage systems capable of draining flammable liquids are sometimes required. Many countries do not require such measures, holding basins in particular; while others prefer to use the systems used to drain surface water.

❖ Detailed comparison

4.2.3.3 Comparison tables

- S31 Gradient along tunnel

Country	Ref.	Requirement	Comment
Italy		No reference.	
Switzerland		No reference.	
Germany	**[1,4]**	Tunnels should have a uni-directional gradient (falling or rising, not changing between falling and rising) greater than the rolling resistance of trains. (This requirement is not designed for drainage of flammable liquids but probably enables a train without power supply to roll out of the tunnel.)	
France		No reference.	
U.K.		No reference.	
U.S.A.		No reference.	

- S32 Track drainage system (drainage and basin)

Country	Ref.	Requirement	Comment
Italy	[1]	Not mentioned in relation to flammable liquids.	
Switzerland	[3]	The track drainage system (in general a combined system) is equipped with a stopping device at the portal and leads to a holding basin (MN11). It is recommended that specific cases are checked for NT considering the local situation (e.g. contamination of ground or surface water)	
Germany	**[1]**	Not mentioned in relation to flammable liquids.	
France	**[1]**	(4.1.1) Drainage system in tunnels with length over 5km on lines with transport of dangerous goods. Basin for 80m ³ released liquid.	
U.K.	**[1]**	Not mentioned in relation to flammable liquids. (56.c) Adequate and reliable means of draining any reasonably foreseeable leakage of water should be provided. The drainage capacity should also take into account the amount of water likely to be used in fire fighting	
U.S.A.		No reference.	
UNECE	**[10]**	(C.2 06) Track drainage system of the appropriate dimensions is safety and environment protection measure. The system should be designed to remove ground water infiltrating through the lining, snow or rain brought into the tunnel by trains, spillage from bulk liquids in transit or fire-fighting water. It is suggested that there should also be a retention basin. This is not an essential measure for passenger- only tunnels but is highly recommended for freight traffic, especially if dangerous goods are frequently transported. The retention basin could be used to retain polluted spillage or fire-fighting water for appropriate disposal without environmental damage. If this basin is enclosed, the risk of fire or explosion should be considered.	

4.2.4 S4 – Open areas

4.2.4.1 Role of the measure

Areas can be specifically set aside for use by the emergency services, to provide standing and access for rescue vehicles, triage and first-aid areas, helicopter landing areas and ready access to tunnel entrances. This minimises the difficulty in reaching the incident scene and evacuating casualties, and allows specialist equipment to be positioned as needed.

4.2.4.2 Synthesis – comments

The provision of open areas near tunnel portals is seen as an important measure needed everywhere. No major differences among the national regulations/guidelines have been identified.

4.2.4.3 Comparison tables

- S41 Rescue areas.

Country	Ref.	Requirement	Comment
Italy	[1]	[3.2.3]…*Emergency rescue area:* illuminated, with a surface of not less than 500m². *Street-level area:* not less than 20m in length, for placing a bimodal vehicle on the tracks, as well as to allow the possibility of crossing the tracks with road vehicles. A raised water tank with a storage capacity of at least 40m³ shall be provided in the area. Assessment shall also be made of the possibility of connecting the water reservoir to a permanent water main. [3.2.4]….*Helicopter landing areas*: it is necessary to identify a suitable area for a helicopter landing site near the main entrances, where a landing area can be built that is easily reached by rescue vehicles situated in the emergency lay-by. The instructions in the M.D. of March 10, 1988, published in the Official Gazette no. 205 of 1/9/1988 and concerning unmarked aircraft surfaces, shall be followed for all provisions inherent to the emergency rescue areas. [3.2.5]… *Triage area*: the creation of an area, having a total surface of not less than 500 m², for emergency medical treatment is desirable near the tunnel entrances, where an area for first-aid and handling the injured can be created.	
Switzerland	[3]	- Transfer points that can be reached by cars or helicopters should be in place. - Areas used to store matériel for rescue purposes should be in place near tunnel portals.	
Germany	**[1,4]**	Rescue areas should be provided in accordance with DIN 14090 and must have a total area of at least 1,500 m2. At tunnel portals, rescue areas should be set up at top-of-rail level. In those cases where such a rescue area is not suitable for a rescue helicopter to land; possible landing sites should be designated in the immediate vicinity. Dividing up the required total area of the rescue area into several smaller areas is permitted if this gives a reduction in the distance to the tunnel portal or emergency exit.	
France	[1]	(3.1.1) - …parking areas that allow for the stopping and reversing of rescue vehicles; - …areas for helicopter landing near the tunnel entrance.	
U.K.	[1]	Emergency access provided at the tunnel portals and other access	

❖ Detailed comparison

Country	Ref.	Requirement	Comment
		points should include: • adequate access from the highway for vehicles and pedestrians; • hard standing for emergency vehicles.	
U.S.A.		No reference.	
UNECE	[10]	- (C.4 03) …Where possible, an area (ca. 500 m2) with road access should be reserved for emergency services vehicles….. ..at both portals and at any emergency exits. ….emergency services areas including the access roads and passing places should have a suitable all-weather surface able to support the vehicles likely to use it. If dual-mode (road-rail) vehicles are to be used in an emergency, a ramp suitable for mode changing should be installed adjacent to each portal. ….a helicopter landing area should be provided additional to the area provided for the emergency services. - (C.4 04)…. Modification of the railway track to make it suitable for road vehicles is only recommended if the use of road vehicles inside the tunnel is part of a comprehensive intervention and rescue concept based on the fire brigades plan.	

4.3 Safety equipment

4.3.1 E1 - Smoke control ventilation

4.3.1.1 Role of the measure

Under normal operating conditions, a ventilation system is generally not necessary. This is a clear difference from road tunnels. Rail tunnels typically have a higher blockage ratio than road tunnels, and moving trains have a higher impact on airflow than moving road vehicles. Thus higher efficiency ventilation systems are needed in rail tunnels. Also, these are only likely to be effective once all train movement within the tunnel has stopped, because all trains have either come to a halt or have left the tunnel.

It is possible to identify three situations where ventilation might be needed:

a) In the incident tunnel: mechanical smoke extraction system in the main running tunnel to draw out smoke or create a defined airflow to keep one side of the fire smoke-free for rescue and fire fighting.
b) Where twin tubes are used, to confine smoke to the incident tube, keeping the unaffected tube free of smoke and preventing cross flows disturbing smoke extraction.
c) To maintain safe areas in emergency exits, cross-passages or a parallel escape tunnel free from smoke (by producing an overpressure within the safe areas).

For railway applications, the positive effects of longitudinal ventilation are a matter of debate, even controversy. The situation cannot be compared to road tunnels, where different ventilation systems, e.g. transverse ventilation, can be installed relatively easily in longer tunnels. In road tunnels, the cost-benefit of complex ventilation systems is different, as such systems are an operational necessity to maintain air quality in the tunnel, where in many rail tunnels a ventilation system is eventually only needed in an emergency.

4.3.1.2 Synthesis – comments

The aspects that must be taken into consideration for parameters E11 and E12 are some of the most important and controversial ones. This is in part due to the sizeable economic costs associated with installing jet fans (the entire tunnel needs to be made larger to accommodate the fans). Also there is some debate over the practical effectiveness of the longitudinal ventilation of main tunnels in the event of a fire. The difficulty of controlling air flow, in terms of both space (number and position of the Jet fans simultaneously in operation) and time (the delay in detecting a fire and then setting the entire ventilation system in motion), means that unquestioning belief in the effectiveness of longitudinal ventilation may be ill-founded.

Analysis of those regulations taken into consideration reveals two opposing conclusions: on one hand, Italy, Switzerland, Austria and Germany substantially agree that minimal advantage, and even major risks, can be derived from forced longitudinal ventilation and maintain that natural ventilation (due solely to pressure differences between the tunnel ends or induced by the openings to the surface) combined with the piston effect from moving trains are sufficient to meet safety demands and are economically advantageous, (there are no explicit reference regulations for Germany, but recent constructions, for example the new Colone-Frankfurt link, give a clear indication of the current thinking); on the other hand, France and the UK have regulations referring to the adoption of forced, longitudinal ventilation systems. More specifically, the French regulations require the compulsory establishment of permanent overpressure conditions in major tunnels, and for all mixed-traffic tunnels over 5,000 m long.

4.3.1.3 Comparison tables

- E11 Natural ventilation

Country	Ref.	Requirement	Comment
Italy	[1]	(Recent developments 2001)… In double-tube tunnels, it is deemed opportune not to establish permanent overpressure conditions in main tunnels, but to maintain cross-connections in constant overpressure, so as to prevent fumes from spreading into the tunnel constituting the “safe place”. Hence, it is deemed opportune <u>not</u> to establish permanent longitudinal ventilation in main tunnels.	
Switzerland		No reference.	
Germany	[1]	Ventilation of main tunnels not mentioned, the principle of using natural ventilation is implicitly agreed.	
France	[1]	See E12.	
U.K.		See E12.	
U.S.A.	[1]	See E12.	
UNECE	[10]	See E12.	

- E12 Forced ventilation

Country	Ref.	Requirement	Comment
Italy	[1]	- (2.2.5) Based on the results of studies conducted internationally (Memorial Tunnel in Baltimore in the U.S., Project Eureka Fire-tun. in Europe), the idea of longitudinal ventilation of tunnels does not seem practicable, since the parameters that come into play are numerous and difficult to manage. In many cases, the effect of this kind of ventilation has not only been ineffective, but even counterproductive, creating a “backlayering” effect. …..the presence of forced ventilation equipment inside tunnels is thus not required, due to its limited effectiveness and the impossibility of it working in the geometry existing tunnels. - (3.1.6)…Appropriate technical solutions should be adopted in order to keep exit passages smoke-free. Suitable measures, pertaining to railway tunnels, must be examined.	
Switzerland	[!]	This aspect is not considered by the regulations listed in chapter 1. However, according to recent information: - in double-tubes, exit passages and cross-connections shall be maintained in overpressure; - in major tunnels, longitudinal ventilation systems have never been foreseen.	[!] Information obtained directly from railway network.
Germany		Smoke control ventilation in the main tube is not mentioned.	
France	[1]	*For tunnel lengths of length between 800m and 5000m* ventilation and smoke-extraction systems shall be assessed. *For tunnel lengths of length over 5000m*, a smoke extraction facility is compulsory in cases of the transit of trains carrying dangerous goods or standard rolling stock. (4.2)…Minimum extraction speed of 1.5m/s. Ventilators resistant up to 200°C. Two independent motors.	
U.K.	[1,2, 3]	- [1] The ventilation system should: achieve an acceptable environment under normal operating conditions; control the movement of smoke in case of emergency. If several trains are in the tunnel simultaneously: - steps must be taken to keep other trains from being assailed by smoke; - evacuation procedures must be identified. - [2,3] The decision …..and the corresponding performance requirements for such a system should result from a risk assessment.	
U.S.A.	[1]	(3.2.4.3) A ventilation system for the contaminated tunnel shall be designed to control smoke in the vicinity of the passengers. (4-3.1) Such system is designated for use in fire emergencies .. in either the supply or exhaust mode. Fan motors…designed to achieve their full operating speed in no more than 30..- 60 seconds.	
UNECE		(C.2 05) The assessment of the airflow in a tunnel should consider tunnel and train aerodynamics, the fresh air supply (for physiological needs), the control of heat and smoke from a fire and the control of pollution (diesel). Ventilation design should take into account the associated risks and costs. Ventilation systems must be designed to keep emergency exits, cross passages and safety tunnels free of smoke.	

4.3.2 E2 - Emergency exits and rescue access ventilation

4.3.2.1 Role of the measure

The design of emergency exits and rescue access should include provisions to prevent the spread of smoke into the safe areas, cross-connections, parallel tube and escape/rescue accesses.
Beyond the emergency exits, good conditions should be maintained in order to let people wait temporarily in areas free from smoke and/or toxic gases.

4.3.2.2 Synthesis – comments

The requirements for emergency exits are briefly described below.

4.3.2.3 Comparison tables

- E 21 Emergency exit / rescue access ventilation

Country	Ref.	Requirement	Comment
Italy	[1]	(3.1.6)… Appropriate technical solutions must be adopted in order to keep exit passages smoke-free. Suitable measures, pertaining to railway tunnels, must be examined.	
Switzerland		No reference.	
Germany	[1,4]	[1] Smoke control ventilation not mentioned. [4] (3.2.3.4) …Should the minimum length of locks not be achieved, appropriate compensation measures are required when demonstrating the same safety level. At present, only mechanical systems for overpressure ventilation can be considered for this purpose.	
France		No reference.	
U.K.		No reference.	
U.S.A.		No reference.	

4.3.3 E3 – Lighting systems

4.3.3.1 Role of the measure

In case of an emergency it is important to have sufficient lighting in the tunnel. Light is needed for evacuation and rescue operations. In case of a fire, additional directional lights may indicate the route to the exits. Also in the escape routes (cross passages, escape tunnel etc.) sufficient lighting is necessary to allow effective evacuation. Thus, emergency lighting is provided to guide passengers and staff to a safe area, or to the outside once they have entered a lateral/vertical exit, a cross-connection or shaft, in case of an emergency.

❖ Detailed comparison

4.3.3.2 Synthesis – comments

The specific requirements vary significantly among the different guidelines.

4.3.3.3 Comparison tables

- E31 Emergency tunnel lighting

Country	Ref.	Requirement	Comment
Italy	**[1]**	(1.2.2/ 2.2.2/ 3.1.9) •*Height of light*: Height ≤ 2m over the walkways, and directed onto them. •*Location*: Single track tunnel: one side (same as walkway); Double track tunnel: both sides. •*Luminosity*: Luminance of 5 lux at a height of 100 cm over the walkway surface. •*Safety concept*: Lighting is off during normal operation, and is turned on in case of an emergency. The emergency lightning in the tunnel is turned on by a permanently lit emergency (push)-button. •*Other lighting*: In some tunnels niches and refuges are permanently lit. Niches and refuges can also be equipped with portable lamps to be used in case of an emergency. •*Autonomy and reliability*: Independent electrical circuits are required for emergency lighting. Cables must be protected against water and heat through conduits or fireproof ducts. Cables connecting the devices to the protected cables must have fireproof insulation (CEI 20-36). Cables must run from both tunnel portals to provide a redundant supply should one circuit be damaged.	
Switzerland	[3]	Emergency lighting: generally as floodlights every 50m on one tunnel side, 80cm above escape route. Switches in tunnel to turn on lights. For Existing Tunnels: An evaluation must be made based on a tunnel-specific risk assessment and cost effectiveness criteria. (in general this means lighting for all new tunnels and for existing tunnels with a length over ~ 0.5 – 1km	Info derived from first issue of this report nov 02.
Germany	**[1,4]**	Escape lighting, including illuminated exits, must be installed on both sides of tunnels longer than 1000m, at a height of at least 2.50m. Technical specifications based on DIN 5035 (harmonised with EN 1838). • *Illumination*: min. . 1 lux. • *Luminosity*: …minimum luminance of 0.5 lux and uniformity of at least 1:40 (DIN 5035, Part 5) must be ensured. It must be possible to turn on the emergency lighting from the operation centre, each tunnel portal and from inside the tunnel (every 125m, 250m from the portal). Switches installed within the tunnel must only switch the lights on, "off" switches are not permitted within the tunnel. •*Autonomy and reliability:* Guaranteed emergency power supply for at least 3 hours	
France	[1,!]	- [!] Escape lighting must be installed both-sides in tunnels >1.000 m including the exits. Technical specification based DIN 5035 (harmonised with EN 1838). Illumination about min 1 lux. Switches installed within the tunnel should be at ~125m separation, and must only switch the lights on. "Off" switches are not permitted within the tunnel. In single or double-track tunnels, lights should be on both sides of the tunnel in staggered rows. General lighting (also for maintenance) from fittings every 25 m on each tunnel wall, staggered, giving 2 lux on walkways. These are normally switched on. In some tunnels there are two categories of light; escape light and normal light. During normal operations the escape lighting is switched off and is only switched on in case of emergency. - (3.2.2)…Emergency lighting of 'B' type, providing at least 2 lux at ground level for 1 hour. The emergency lights are powered in such a way that no fire can cause them to fail over a tunnel section longer than 100m. The maximum distance between two lights is 50m. In double track tunnels on both sides with alternating arrangements	[[!] Information obtained directly from railway network.

Country	Ref.	Requirement	Comment
U.K.	[1]	(59.a) Running tunnels, cross-passages and access shafts should be permanently equipped with adequate lighting. …In the event of a total power failure, it should be possible to sustain emergency lighting at not less than 5 Lux for at least the time required for evacuation and not less than 3 hours.	
U.S.A.	**[1]**	-(3-2.4.7.1.1) Emergency lighting systems…in accordance with NFPA 70… -(3-2.4.7.1.2) Exit lights, essential signs, and emergency lights shall be included in the emergency lighting system… -(3-2.4.7.1.3) The illumination levels ….shall not be less than 0.25 ft-candles (2.69 lux) at the walking surface.	
Other Spain		Escape lighting and normal lighting must be installed on both sides of the tunnel, including at the exits. "Normal" lighting is switched on during normal operations. Every 50 m on each tunnel wall, staggered to give 25 m between lights.	
Other Finland		A safety lighting and sign illumination system as well as emergency exit signs should be located in railway tunnels to ensure that people can get out and to safeguard rescue operations (1 to 3 h). The power supply to the system must be through battery sets. The system must be designed and built in accordance with the set of guidelines 147/01/87 issued by the Rescue Department of the Ministry of the Interior, and [the Standards] SFS 4640/1985-06-03 and SFS EN-60598-2-22. The sign illumination must be switched on permanently during each operating. The safety lighting must provide at least 1 lux at walkway level in the running tunnel, and at least 2 lux in areas difficult to negotiate, such as staircases. Arrangements must be in place for external control of the safety lighting and for the monitoring of the battery sets. The operating principles of the system should be selected separately for each individual location (lighting unit/central accumulator system).	
Other Sweden		Basic standard: Light from fittings so that the light is at least 3 lux between the lights. Normally switched off. To be switched on remotely. Additional standard: Guaranteed power supply for emergency or alternative system to ensure high reliability. Supply cables protected against mechanical impact and fire. Power supply/lighting should be supplied in sections.	
Other Norway		Escape lighting shall be installed in all tunnels categorised A, B and C. The escape lighting shall provide: • light during the evacuation period • sufficient light for a safe evacuation Light fittings are installed ca. 0.65m above the rail top. Maximum distance between two lamps is 10 m. The escape lighting must function in smoke-free and smoke conditions. The technical specification is based on the new ISO-standard ISO/DIS 16069, which is not yet approved.	
Other Denmark		Escape signs, exit signs and standard signs have clear bottom, so that light is thrown down on the walkway. Common light (also for maintenance) from fittings every 25 m on each tunnel wall, staggered, gives 10 lux on walkways. Normally switched off. Switched on remotely.	
UNECE	**[10]**	(C.3 04) Emergency tunnel lighting shall be installed on one or both sides of the tunnel, especially in tunnels used by passenger trains. Escape walkways shall also be properly lit. Emergency lighting shall be reliable and operating under autonomous conditions, visible under smoke and other poor visibility conditions.	

- E32 Emergency exit / rescue access lighting

Country	Ref.	Requirement	Comment
Italy	**[1]**	Same used in the main tube, see E31	
Switzerland	[3]	Not explicitly mentioned, see E31	
Germany	[4]	Tunnels and emergency exits must be provided with emergency lighting as safety lighting according to DIN 5035, Part 5, and VDE 0108. In the event of a loss of power or short circuit in the external supply line, the emergency lighting must provide the required illumination for a minimum of at least 3 hours.	
France	[1]	See E31	
U.K.	**[1]**	See E31	
U.S.A.	**[1]**	See E31	

4.3.4 E4- Escape signs in tunnels

4.3.4.1 Role of the measure

This evacuation and rescue measure generally indicates the direction and distance to a safe place. All signs should meet current European standards for safety signs. Emergency exit signs are mostly produced with white text on a green background (CEN or ISO norm). Escape signs are installed on the side walls (at different heights) and indicate the distances to the nearest exits in each direction. The location of emergency equipment may also be shown on signs.

4.3.4.2 Synthesis – comments

Signage is described in all guidelines, but some requirements differ from one national regulation/guideline to another.

4.3.4.3 Comparison tables

- E41 Pedestrian exit signs

Country	Ref.	Requirement	Comment
Italy	[1]	(3.3.3) …Escape signs can be either reflective or luminescent. Fire reaction category: 0 •*Shape:* Square or rectangular. •*Pictograms:* White on green background (green colour must cover at least the 50% of the surface). •*Location:* Escape signs should be placed every 100m, and indicate the closest escape exits or the distance to the tunnel portal. Luminescent signs close to emergency exits show "emergency exit" written in Italian and English.	•Peculiarity: Reflective signs can only be seen if directly illuminated; if illuminated signs are present these are switched off and lit only in case of an emergency. Signs are typically placed adjacent to niches, where they are visible due to the alcove lights.
Switzerland	[3]	Characteristics: • *Dimensions:* 0.60x0.20m; • *Distance*: every 100m; • *Colours*: green sign; • reflective; • Mounted on walls, over escape passages, at the height of the tunnel spotlights. •*Materials:* Aluminium, covered by protective fire-proof film, and/or materials having similar properties.	
Germany	[4]	Marking is provided along the rescue routes in the main tunnel only. The escape-route marking comprises direction arrows and rescue signs. There are two kinds of rescue signs - those that indicate the distance to the nearest safe place, and those that identify an emergency exit. Separation of ca. 125m (to be located at light switches). Indication of preferred escape direction and distances to exits in both directions. Additional arrows at distances of ca. 25m, indicating the nearest exit. Signs must be reflective and luminescent.	
France	[1]	(3.2.3) …Indication of preferred escape direction and distance to the nearest station or exit every 100 m.	
U.K.	[1,2,3]	- [1] (51.f) (52.c) (53.e) Suitable signs should be provided to indicate the direction and distance to the adjacent emergency access points or cross-passageways. - [2,3] Illuminated and uniquely identified signage, powered from the emergency lighting system should identify any lateral points of evacuation from the tunnel.	
U.S.A.	[1]		
UNECE	[1]	(C.3.03)…Tunnels should be marked with standard signs pictograms. Signs should be fixed in the tunnel to indicate the direction and distance to any safety feature such as: exits, cross passages, telephones, etc. Signs should indicate the emergency equipment available to passengers and other potential users. The	

❖ Detailed comparison

Country	Ref.	Requirement	Comment
		signs indicating “Emergency exits” should conform to the pictograms proposed by the ISO 6309 standard or the CEN norm of December 2000. The background colour should be green.	
Other Spain		Escape signs to be located at lighting points. Indication of both escape directions must be provided.	
Other Finland		Signs for exit routes and emergency telephones must be located in each tunnel at intervals of one hundred (100) metres close to emergency lights. The signs should indicate the walking distance to the beginning of the compartmentalised exit route or to the open air. The signs must be made in accordance with ISO Standard 3864 and the Signalling Systems by the RHK (the Finnish Rail Administration Board) /55*/.	
Other Sweden		Basic standard: Escape signs on both tunnel walls every 100 m with information about distance to safety and distance to alarm telephone. These are permanently lit and they indicate direction and distance to nearest exits (cross passages). Exit signs at exits: The signs have white text on green background.	
Other Norway		Escape signs in all tunnels categorised A, B and C indicating direction and distance to the two nearest exits. The signs shall indicate: • direction and distance to nearest exits • tunnel gradient • direction name, A and B. The signs shall be photo luminescent.	
Other Denmark		Escape signs on both tunnel walls every 62 m. These are permanently lit and indicate direction and distance to nearest exits (eg. cross passages). “Exit” signs at exits: The signs have white text on green background. Standard signs, white, permanently lit, with SOS (on blue background), telephone and extinguisher at every exit. Information signs at every exit, on far side, indicating the name of the tunnel and the distance to the tunnel portals (only Great Belt). Number of the exit to be seen by the engine driver while passing through the tunnel	

- E42 Other

Country	Ref.	Requirement	Comment
Italy	[1]	- Signs with white pictograms on red background are used to indicate the location of emergency equipment (telephones, extinguishers etc.). - See E41.	
Switzerland		No reference.	
Germany		No reference.	
France		No reference.	
U.K.		No reference.	
U.S.A.		No reference.	

4.3.5 E5 - Communication and alarm system

4.3.5.1 Role of the measure

It is important for the Train Control Centre to have all of the relevant information if an emergency situation arises (including the presence of a fire). Such information is achieved by surveillance of the tunnel and through various communication systems. Communication can be automatic, using systems triggered by accident detection equipment or it can be manual, e.g. by alarm push-buttons or emergency telephone. Onboard and line-side detection systems are also crucial in increasing the likelihood of detecting an emergency situation/fire and applying the appropriate intervention strategy. Furthermore, the communication system is used to instruct tunnel users about what do. Announcements may be made by radio, through emergency telephones or, in some cases, through loudspeakers in the tunnel or in refuges. Emergency telephones or some similar means of communication, directly connected to an operation centre (independent of train radio or mobile phone) can be available so that passengers, too, can use them in emergencies. Emergency telephones permit adequate and reliable communication during emergencies.

4.3.5.2 Synthesis – comments

Emergency telephones are available in nearly all tunnels, but the distance between telephones and other detailed arrangements, vary. As a rule, there is an alarm in the Train Control Centre when safety equipment is used and radio coverage in the tunnel. GSM-R may also provide a further mitigation measure. A general synthesis of the national guidelines cannot be drawn up.

4.3.5.3 Comparison tables

- E51 Emergency telephones

Country	Ref.	Requirement	Comment
Italy	**[1]**	(1.2.4/2.2.4/3.3.5) - Railway telephones: telephones at portals and every 500m inside tunnels. - Emergency phones for the public: connected with the operation centre, every 250m (same installation can be used by personnel for loudspeaker announcements inside the tunnel).	Where GSM-R is present the installation of emergency telephones might not be necessary
Switzerland	[3]	Emergency telephones placed in specific points.	
Germany	**[4]**	Tunnels must be equipped with emergency telephones, placed: at the tunnel portals; within the main tunnel; in the immediate vicinity of the emergency exits; within emergency exit routes inside the external doors. Locating emergency telephones within the locks is not permitted. In double-track tunnels, the emergency telephones should be provided opposite each other on both tunnel walls. Marking of phones according VBG 125. Ground-train radio and fixed emergency telephones are available for exchanging information during self-rescue.	

❖ Detailed comparison

France	[1]	(3.2.5)…Two separate lines should be provided, at least, for management of rescue operations and fire-fighting water supply operations.	
U.K.	[1]	(61) Telephones connected directly to the railway control should be provided at appropriate intervals and in suitable locations including at any cross-passages and access points.	
U.S.A.	[1]	- (8-4.1) The system shall have a telephone network of fixed telephone lines and handsets… - (8-4.2) …The location and spacing of telephones along the train-way shall be determined by the authority having jurisdiction. Telephones along the train-way shall have distinctive signs and/or lights for identification.	See [1] (3-1.5.)
UNECE	[10]	(C.3 05) …emergency telephones should always be installed at the key points in tunnels – cross passages, on escape walkways and shafts. Telephones should be able to function properly and work in the tunnel environment with a potentially high noise and poor light. ….they should be installed with a sound hood to avoid noise-affecting conversation. Emergency telephones should be linked to the emergency centre in the railway operations control centre. Emergency telephones should not be linked directly to fire or other rescue services. If the direct radio or GSM-R link between the train and the operations centre exists, installation of emergency telephones might not be necessary.	

- E52 Alarm push-button (manual fire alarm)

Country	Ref.	Requirement	Comment
Italy	[1]	When an emergency push-button is activated, this must connect directly to the train operating centre. On pressing the emergency button the emergency lighting will be switched on and the train control centre should be able to identify the location of the emergency “viva-voice” device. The emergency (push)-button locations must be permanently lit.	[!] Information obtained directly from railway network.
Switzerland	[2]	Suggested but not prescribed	
Germany	[4]	The connection to the operations-monitoring office must be set up by pressing an emergency button and without any further operations. Pressing the emergency button must set off an acoustic signal in the operations-monitoring office and the location of the telephone must be transmitted automatically.	
France		No reference.	
U.K.		No reference.	
U.S.A.		No reference.	

- E53 Fire/smoke detection

Country	Ref.	Requirement	Comment
Italy	[!]	Those niches containing electrical and signalling closets must be equipped with smoke and heat detectors in order to identify any possible ignition of a fire.	[!] Information obtained directly from railway network.
Switzerland		No reference.	
Germany	**[4]**	Measure dismissed, as it is not regarded as being effective	
France		No reference.	
U.K.	[2,3]	(c.11) The infrastructure controller shall carry out a risk assessment to determine whether any surveillance or detection systems are required in the new or re-opened tunnel so as to allow the safe operation of the tunnel under normal, degraded and emergency operating conditions.	
U.S.A.	[1]	(3-2.7.1) Heat and smoke detectors installed at traction power substations and connected to the central supervising station.	

- E54 Radio rebroadcast

Country	Ref.	Requirement	Comment
Italy	**[1]**	Tunnels and service/lateral tunnels must be equipped with radio communication systems to allow communications between Fire Brigade and Railway emergency team.	
Switzerland	[3]	Ensure radio communication between emergency and railway personnel on fire fighting train (compatible sets) (MN23). Standard for all tunnels.	
Germany	[1,4]	The normal radio system used by the rescue services must be available within a tunnel without any restrictions. This also applies to the necessary radio connections between the site of the operation and the operations command centre (BOS radio installations)	
France	[1]	Not explicitly mentioned (3.2.1)…Connection between train and operation centre must be assured.	
U.K.	[1]	(60) A radio communication network should be provided. Discrete radio between: • train drivers and train control centre (TCC); • TCC and the public address to passengers on a train; • an 'open' radio between TCC and all trains simultaneously, including public address to passengers in them.	
U.S.A.	[1]	(8-3.1) …passenger rail system shall have at least one radio network that is capable of two-way communication with personnel on trains, motor vehicles, and all locations of the system. (8-3.2) Wherever necessary ….a separate radio network capable of two-way radio communication for fire department personnel to the fire department communication centre shall be provided.	
UNECE	[10]	(C.4 07) Radio continuity should be provided for the fire and rescue services linking fire fighters with their immediate command to ensure operational efficiency during an emergency. The system must be reliable and allow the rescue services to use their own communication equipment when needed.	

❖ Detailed comparison

- E55 Loudspeakers

Country	Ref.	Requirement	Comment
Italy	[1]	(1.2.4/ 2.2.4/ 3.3.5) ….As a rule, on the most important trains, there is an information system for passengers …that is used by on-board railway personnel to make the necessary announcements. However, …for managing emergencies, it is deemed necessary to equip tunnels with an effective internal loudspeaker system that shall be used, when necessary, by railway personnel or rescue teams.	
Switzerland		No reference.	
Germany	[1,4]	[1] No reference. [4] It must be possible to make loudspeaker announcements only on passenger trains. Loudspeaker announcements offer support to the self-rescue measures since the train staff can give instructions.	
France		No reference.	
U.K.		No reference.	
U.S.A.		No reference.	

4.3.6 E6 - Operation and traffic management

4.3.6.1 Role of the measure

Traffic regulation and management is mainly a preventive measure. Monitoring traffic, its speed and intensity makes it possible to take any actions necessary to mitigate possible consequences of an accident.

4.3.6.2 Synthesis – comments

An optimised timetable is a widely used measure to prevent trains (especially passenger and freight trains) from crossing each other in tunnels. No other general synthesis can be concluded.

4.3.6.3 Comparison tables

- E61 Speed and traffic density monitoring

Country	Ref.	Requirement	Comment
Italy		No reference.	
Switzerland	[!]	All the precautions must be taken, such as fire protection measures for the rolling stock, line side hot box detectors to stop a train before entering the tunnel. There are no general operational restrictions, but in some tunnels there is a timetable separation between passenger and freight trains.	[!] Information obtained directly from railway network.
Germany	[1,4]	[1] Not mentioned [4] The railway infrastructure manager is required to set up the technical conditions and to issue working instructions to ensure that: a train on which the actuation of the emergency brake has been reported can leave the tunnel as quickly as possible; after the train comes to a halt, its location can be determined by the operations-monitoring office even without information from the train staff.	
France	[1]	(3.3.2) …Speed monitoring system to prevent collisions caused by driver error.	
U.K.	[1]	No reference.	
U.S.A.	[1]	No reference.	
UNECE	[10]	(C.1 02) …the signalling system …to prevent one train colliding with another train …It will include any train monitoring or protection system intended to prevent a train from passing a signal set at danger or exceeding a speed limit.	

- E62 Traffic typology / regulation

Country	Ref.	Requirement	Comment
Italy	[!]	No reference, …RFI has stated an operational speed limit of v≤160 km/h in case of crossing between passenger and dangerous goods trains, only on the High Speed line Rome - Florence. It has been also stated that no crossing between passenger and dangerous good trains will be allowed on the new High Speed Bologna – Florence.	[!] Information obtained directly from railway network.
Switzerland	[!]	See E61.	[!] Information obtained directly from railway network.
Germany	[1]	The passing of goods trains and passenger trains is prohibited in single-tube tunnels.	
France	[1]	The passing of trains carrying dangerous goods and passenger trains is prohibited in single-tube tunnels having a length of L > 5000m, but accepted in shorter, single-tube tunnels.	
U.K.		No reference.	
U.S.A.		No reference.	

UNECE	[10]	(C. 1.09) The scenario of a passenger train colliding with a freight train might be avoided if these trains are not allowed in a double track tunnel at the same time. This …is not recommended as a standard measure except for tunnels, which are very long or have mixed passenger and freight trains with dangerous goods. … total separation of traffic may not be necessary if an optimised timetable could prevent passenger and freight trains with dangerous goods from passing through a tunnel at the same time. …Very frequent traffic through particular tunnels could be made safer by separation of operations of passenger and freight trains with dangerous goods into day and night.	

- E63 Tracking status of the train before entering a tunnel

Country	Ref.	Requirement	Comment
Italy	[!]	Systems for detecting temperatures on a running train, by means of thermographical portals are current being trialled experimentally. These systems will be used to prevent accidents in long tunnels. The thermographical portals detects the very early stages of a fire, or other situations with an high degree of risk (out of gauge loads, unbalanced loading, temperature anomalies). The portal is a diagnostic device which can be integrated with the signalling system and the shut-down system of the railroad. If a dangerous situation is detected, the combined system can stop a running train before it enters the tunnel.	[!] Information obtained directly from railway network.
Switzerland		No reference.	
Germany		No reference.	
France		No reference.	
U.K.		No reference.	
U.S.A.		No reference.	
UNECE	[10]	(C.1 03) …Line side detectors of vehicle faults (hot box detectors, etc.) should be installed at a sufficient distance from the tunnel portal such that a defective train may be stopped by the signals before entering the tunnel in order to reduce the risk of an incident in the tunnel.	

4.3.7 E7 – Power supply

4.3.7.1 Role of the measure

In case of an emergency, especially a fire, it is necessary to have a reliable supply of power to operate the safety systems such as lighting, information and communication systems etc. For this reason, an emergency power supply or alternative scheme should ensure the availability of power for a certain period of time.

4.3.7.2 Synthesis – comments

All of the guidelines prescribe long-lasting power supply in tunnels.

4.3.7.3 Comparison tables

- E71 Power supply

Country	Ref.	Requirement	Comment
Italy	**[1]**	*Location:* •(1.2.7) …Inside the tunnel: electrical supply in niches. •Outside: electrical supply in rescue area *Main features:* •(3.1.8) … Each emergency facility …must be linked to the normal power-distribution network and to a source of emergency power that may be composed by: a) a storage batteries with automatic recharging and inverters …autonomy may not be less than two hours, if the system is not coupled to a generating set; b) self-starting generating set. The batteries and generating sets … located in areas not subject to fire risks and suitably ventilated. Any walls adjacent to tunnels must be built with fire-resistant structures of at least R.E.I. 120'. • …emergency electrical power may be installed, using wiring that is independent of the primary system, under the conditions that: - the two power sources are physically remote and under such conditions that, in the event of an emergency, they are not engaged simultaneously; - each system alone should be able to power-up the entire facility.	
Switzerland	[3]	Electrical power must be supplied, if possible, from both entrances. The system shall also be protected to at least IP 64 standard. In long tunnels, the emergency lighting, in case of fire, must be subdivided into sections of max. 500m, in which measures for electrical protection shall be respected.	
Germany	**[4]**	In tunnels > 1km an electrical breaker shall be located every 125m on both sides of tunnel at the same positions as switches for emergency lighting. It must be possible to switch off the electrical supply for trains in the whole tunnel, actuated by the operation centre as well as through switches in the tunnel. There must be indicators at the portals to show whether the power supply is switched on or off. Cables and plug connections are to be laid in such a manner that they cannot be damaged as a result of an accident. The connections must comply with the usual plug connections used by the rescue services. The supply points to be provided in tunnels should have connections for the operation of electrical equipment with a voltage of 230 V as well as 400 V. The operating power shall be at least 8 kW.	
France	[1]	(3.2.1.)…Electric equipment must be connected either to a 1 hour battery or a second independent main (provided with fire proof cables etc.). Electrical supply on both sides of the tunnel (12kVA; 240/400 V) every 200m with redundant power supply	
U.K.	[1]	- [1]…Cable routes should be positioned to minimize damage from any derailed train. - (54.b) If conductor rails are used, they should be ... insulated or shielded to prevent accidental contact by people and their tools. - (54.c) Means should be provided throughout the tunnels…. for the disconnection of traction current. In the case of conductor rail systems, adequate means of instantaneous discharge of current should be provided. The same system of discharging the current should be used throughout.	

U.S.A.	[1]	- (3-2.4.7.1.2) Emergency fixtures, exit lights, and signs shall be wired separately from emergency distribution panels. - (4-7.1) The power for the emergency ventilation fan plants shall originate from two separate and distinct utility sources. - (4-7.2) All wiring materials and installations shall conform to the requirements of NFPA 70.	
UNECE	[10]	(C.4 06) The electricity power distribution system in the tunnel should be suitable for emergency/rescue services' equipment. Standard socket outlets with residual current circuit breakers should be installed. All power outlets for rescue services should be regularly maintained and checked.	

4.3.8 E8 – Fire suppression

4.3.8.1 Role of the measure

In the event of a fire in a tunnel, it is best to fight the fire during its early stages. First-aid and fire fighting equipment on board a train or in the tunnel can be used. In case this equipment is not sufficient, there should be facilities for fighting the fire using external assistance from a fire brigade or similar.

4.3.8.2 Synthesis – comments

The provision of a water supply for use by the emergency services to fight fires is the most widely used measure. This is achieved through a continuous water main into the tunnel (either permanently filled or dry pipe), or through branch pipes to tunnel entrances (portals, emergency exits). In some countries, automatic or manually triggered fire-extinguishing systems are installed in plant rooms, in order to fight a fire at an early stage.

4.3.8.3 Comparison tables

- E81 First response and fire fighting

Country	Ref.	Requirement	Comment
Italy	**[1]**	(3.1.7.) The water fire fighting system comprises dry, steel pipes (fig.6), placed at the base of the abutment and protected so as to guarantee a minimum resistance to fire (REI 60), and nozzle connections, UNI 45, located every 250m. Water storage is located at each tunnel entrance near to the portal. Min Flow rate of water: 200 l/min at minimum pressure of 2bar. Capacity 40 m^3.	
Switzerland		Water supply at tunnel entrance (in order to supply the fire fighting train with water) (MN22). Standard for New Tunnels (giving synergy with facilities required during construction). Water pipe in the tunnel (dry or filled) (MN21). Check valves are not recommended in general, but may be installed in special cases (e.g. water line between two communities through the tunnel).	Info derived from first issue of this report nov 02.
Germany	**[1,4]**	Water supply required at tunnel entrance with safe areas and emergency exits, with a reserve of 96m³, capable of supplying 800 l/min. In each tunnel tube permanent water main along the whole tunnel length, fed through emergency exits and tunnel portals, connected at cross passages. Hydrants every 125m, static pressure 8 bar, 5 bar if in use, marked through signs.	

France	**[1]**	- (3.2.4.1)…Urban tunnels: water supply via a 0.1m pipe (either dry or protected against frost). Hydrants provided with 2 X 40 mm and 1 X 65 mm nozzles along the tunnel (max separation of 100m). - (3.2.4.2) Urban tunnels: water storage of 120 m^3. Supply via a pipe protected against frost. Hydrants provided with 2 X 40 mm and 1 X 65 mm nozzles along the tunnel (max separation of 250m). Minimum flow rate of water of $60m^3/hr$, at minimum pressure of 6 bar. - (3.3.3)… Provision of rolling pallet for transport of materials and casualties.	
U.K.	**[1]**	- (56.a) ...fire-fighting main …with hydrant points at least at each end of cross-passages and the lobbies of intermediate shafts … locations and intervals … may be determined in consultation with the local Fire Authority. - (56.b) The… main should provide an adequate flow ….provide a pressure at the hydrant outlets of 4.5 bars ± 0.5 bar. A system of leak prevention and early warning in case of leaks should be provided as appropriate.	
U.S.A.	**[1]**	(3-2.7.2.1) A fire extinguishing system is required. It comprises standpipes and hose systems and storage tanks. Standpipe lines shall be a minimum size of 4 in. (101.6 mm) in diameter.	
UNECE	**[10]**	(C.4.0.5) A fire-fighting water supply should be made available in all tunnels covered by these Recommendations. The tunnel designer should consult the fire brigade about the design of this water supply. The water supply system should be regularly tested and checked.	

- E82 Fire extinguishing systems (in plant/machinery rooms)

Country	Ref.	Requirement	Comment
Italy		No reference.	
Switzerland		Fire extinguishers in tunnels (MN7) Fire detection and extinguishing systems in technical installations (plant rooms) (MN6). Consideration is recommended for new long tunnels (need will depend on specific case) and also for critical stretches of existing long tunnels	Info derived from first issue of this report nov 02.
Germany		Not explicitly mentioned. Should the local rescue services' existing equipment not be sufficient for operations in tunnels, the railway infrastructure manager is required to conclude special agreements with the responsible bodies concerning the necessary supplementary equipment.	
France		No reference.	
U.K.		No reference.	
U.S.A.		No reference.	
UNECE	**[10]**	(C.2.0.4) …extinguishing systems, for technical rooms should be determined depending on the potential causes of fire. An effective fire suppression system in the main tunnel is not generally practical and is not recommended. Technical rooms, especially those containing safety critical equipment should be suitably protected.	

- E83 Other

Country	Ref.	Requirement	Comment
Italy		No reference.	
Switzerland		No reference.	
Germany		No reference.	
France		No reference.	
U.K.		No reference.	
U.S.A.		No reference.	

4.3.9 E9 – General safety equipment

4.3.9.1 Role of the measure

This section provides a general overview of the main equipment installed in running tubes and cross-connections, in order to facilitate a comparison of the different national norms and regulations at a glance.

4.3.9.2 Synthesis – comments

From the point of view of safety and mitigation measures, some differences are noted among the regulations covering the most important safety equipment. Almost all of the regulations provide for the same basic level of fitting out, but not all supply a detailed description of the required fire main, and in only a few cases anticipate longitudinal ventilation systems in tunnels. However, these differences are due to different approaches to the same perceived problem, exemplified by the use or otherwise of ventilation systems in the tunnel.

4.3.9.3 Comparison tables

- E91 General safety equipment in tunnels (running tubes)

Country	Ref.	Requirement	Comment
Italy	[*]	• Systems for keeping escape routes smoke-free • Fire main, dry-pipe • Tunnel lighting during normal operations. • Lighting for escape routes • Emergency lighting (5 lux at 100 cm from the walkway) • Refuges equipped with boxes containing: fire-fighting kits, fire hoses and nozzles, disposable masks, lamps with tripods, electrical wire. • Luminescent and/or illuminated directional signals, indicating escape routes • Loudspeaker system Telecommunications facility	[*] Information derived from the whole of available norms / regulations.

Switzerland	[*]	• Lighting • Alarm buttons • Emergency light switches • Tunnel lighting during normal operations. • Emergency lighting in case of fire • Communication facilities • Handrails and/or touch-guide panels • Reflecting directional signals, indicating escape routes Emergency telephones	[*] Information derived from the whole of available norms / regulations.
Germany	[*]	• Fire main • Emergency lighting (min. 0.5lux) • Reflecting and illuminated directional signals, indicating escape routes • Electrical power outlets (max. dist. betw. connections: 125m) Telecommunications facility and emergency telephones	[*] Information derived from the whole of available norms / regulations.
France	[*]	• Fire main, dry or wet and protected against frost • Lighting • Signs • Ventilation/smoke-exhaust (permanent ventilation in the tubes) • Tunnel lighting during normal operations. • Emergency lighting (B-1H type, 2 lux on the walkways) • Emergency, fire resistant electrical power systems for safety devices • In double-tube configurations, devices to prevent smoke re-entry at portals. • Illuminated directional signals, indicating escape routes (every 100m) • Telecommunication systems (minimum 2 radio lines) Ventilation/smoke-exhaust systems	[*] Information derived from the whole of available norms / regulations.
U.K.	[*]	• Lighting • Alarm buttons • Switches for emergency lighting • Tunnel lighting during normal operations. • Emergency lighting in case of fire • Communication facilities • Handrails and/or touch-guide panels • Reflecting directional signals, indicating escape routes Emergency telephones	[*] Information derived from the whole of available norms / regulations.
Other Austria	[*]	• Emergency lighting (1.0 lux) • Switches for emergency lighting (max. distance between.: 50m) visible in the dark • Direction signs, indicating escape routes (every 25m) and first-aid stations (every 150m) • Electrical power outlets both sides (max. distance between: 125m) • Hydraulic systems for fire-fighting with dry ducts or permanently filled • Emergency telephones Radio transmitters	[*] Information derived from the whole of available norms / regulations.

❖ Detailed comparison

- E92 Cross-connections, general safety equipment

Country	Ref.	Requirement	Comment
Italy	[*]	• Positive pressure systems. • Telephones • Illuminated signs • Emergency lighting Fire and smoke-proof doors fitted with anti-panic handles.	[*] Information derived from the whole of available norms / regulations.
Switzerland	[*]	• Systems for facilitating rescue operations: • Emergency lighting, • Handrails, • Signs, • Telephones, Positive pressure systems.	[*] Information derived from the whole of available norms / regulations.
Germany	[*]	• Fire and smoke-proof doors, closing automatically. Emergency lighting	[*] Information derived from the whole of available norms / regulations.
France	[*]	• Fire and smoke-proof doors (fire resistant up to 2 hrs) at each tunnel end Devices for preventing smoke and fumes spreading into cross-connections and, thus, towards the tunnel not involved in the accident and/or the safety shaft.	[*] Information derived from the whole of available norms / regulations.
U.K.	[*]	In order to establish the equipment needed in cross connections, the following aspects must be considered: • Passage of smoke and heat through cross-connections; • Opening and closing of doors; • Risk to passengers from trains travelling in the other tube (including aerodynamic effects). The most common facilities are: • Ventilation systems for providing fresh air and positive pressure. • Fire and smoke-proof doors Lighting system. The lighting is usually turned off, but it should be of remote operation from adjoining stations and the control room, manual operation from the tunnel and automatically on electrical failure.	[*] Information derived from the whole of available norms / regulations.
Other Austria	[*]	• Fire-proof panic doors, located at tunnel ends. • Positive pressure systems. Emergency telephones.	[*] Information derived from the whole of available norms / regulations.

4.4 Structural & equipment response to fire

4.4.1 R1 – Reaction to, and resistance to, fire

4.4.1.1 Role of the measure

The tunnel structure and the equipment installed in it should be able to resist fire and continue to perform safely during the time necessary for the evacuation of tunnel users. In addition, the structure and equipment should assist fire-fighting, and should be designed with the aim of minimising the economic consequences of a fire.

4.4.1.2 Synthesis – comments

General specifications for the fire resistance of the tunnel structure and equipment are available in all guidelines.

4.4.1.3 Comparison tables

- R11 Reaction to fire

Country	Ref.	Requirement	Comment
Italy	[1]	The materials used for construction or present in the tunnel must have a fire resistance of over 120'. (REI 120) and flammability category 0. Emergency lighting must be designed with fire-resistant wiring (CEI 20-36).	
Switzerland	[!]	Materials used in the tunnel must have a fire-resistance index of over REI 90.	[!] Information obtained directly from railway network.
Germany	[1,4]	Pursuant to the EBA Directive, tunnels and emergency exits must be built of non-combustible materials. A time-temperature profile is given, to form the basis for the composition of the concrete and for the reinforcement	Fire duration [min] 0 5 60 170 Temperature [°C] 0 1200 1200 0
France	[1]	(2.1) All materials must be classified M0 for performance against fire. Category M1 is allowed for lateral linings and light, translucent cover materials (lamps). Fire resistance of the structure and other components should be demonstrated (by calculation or direct testing): • against the normalised time-temperature curve (ISO 834 – EC1); • against the hydrocarbon curve (EC1).	
U.K.	[1]	(50.d) Materials should be chosen to: • resist the spread of flame; • reduce the rate of heat release; • reduce hazardous products of combustion. An assessment of the risk of fire, and the measures that may be taken to minimize the risk, should be made at an early stage.	
U.S.A.	[1]	- (3-2.1.3) Walk surfaces designated for evacuation of passengers shall be constructed of non-combustible materials.	

		- (3-2-5.2.3) Cover board / protective material shall have a flame spread rate of not over 25 when tested in accordance with NFPA 255 (ASTM E 84). - (3-2.5.2.4) …Insulating material for the cable, connecting power to the rail shall meet the requirements of IEEE Standard 383, *Standard for Type Tests of Class I Electric Cables, Field Splices, and Connections for Nuclear Power Generating Stations,* Section 2.5.	
UNECE	[10]	See R12.	

- R12 Fire protection requirements for structures

Country	Ref.	Requirement	Comment
Italy	[1]	See R11.	[!] Information obtained directly from railway network.
Switzerland	[3]	Reference made to materials in general. See R11.	
Germany	[1,4]	In the event of a fire nobody must be injured by spalling of the tunnel lining. The depth of spalling must be estimated taking into account the concrete mixture and reinforcement.	
France	[1]	*(2.2.1) – Tunnels, excavated but not lined:* no special requirement for fire resistance. - *Tunnels with non-load-bearing linings:* design controls should eliminate the risk of secondary collapse (chain reaction). *Other tunnels*: ISO 834 - 2 hrs. - *Tunnels adjoining a structure in a built-up area, reinforced concrete*: 2 hours. - *Tunnels authorised for the carriage of dangerous goods: •* ISO 834 curve: 4 hours; • hydrocarbon curve: 2 hours - *Light linings:* design controls should eliminate the risk of secondary collapse (chain reaction). - *Suspended ceilings and walls that separate ventilation ducts*: 2 hrs, raised to 4 hrs if the carriage of dangerous goods is anticipated	
U.K.	[2,3]	The potential hazard due to materials and equipment falling during a fire should be addressed during the design of a new tunnel, i.e. its lining materials and its reinforcing. Progressive failure modes involving the spalling of small pieces of lining would be less hazardous to fire fighters and others than failure modes which could result in large pieces falling.	
U.S.A.	[1]	(3.2.1.6) Structures such as remote vertical exit shafts and ventilation structures shall be not less than Type I- (332) approved non-combustible construction as defined in NFPA 220.	
UNECE		(C.2 02) ……structural fire protection …especially for those locations involved in any safe haven or rescue. The risk study should consider the likely fire size and its thermal impact on the type of structure involved (heat transfer, smoke leakage, structural damage, spalling, etc.) and the consequences of structural failure. Appropriate temperature development curves should be chosen for the testing of the materials involved.	

- R13 Equipment, resistance to fire

Country	Ref.	Requirement	Comment
Italy	**[1]**	Fire resistant and halogen-free electric cables (according to CEI 20-32). Electrical supplies for emergency systems must use fire resistant cables (CEI 20 -36).	
Switzerland	[3]	(4.3) Electrical supplies to emergency lighting must be fed from both portals, and be fire resistant for 30 min. For existing tunnels, and evaluation of such measures is required, based on tunnel specific risk assessment and cost-effectiveness criteria. Enclosures etc sealed to IP64.	
Germany	[1]	The *structural installations* for emergency lighting, communications, power supply and release of the doors leading out of the emergency exits to the open air, must be designed in such a manner that they can withstand a fire for at least 90 minutes and continue to function (i.e. meet category F 90 according to DIN 4102). Electrical supplies and communication links should be laid in concrete ducts under the escape route wherever possible. The *cables* required for *operations management*, such as signalling or telecommunications cables (except emergency telephones) do not require fire protection. *Locks*: Doors from the lock to the main tunnel must be fire-retardant and smoke-proof (minimum T 30). Doors to a rescue shaft or rescue passage must be smoke-proof.	
France	[1]	- (4.2) Fire resistance of ventilation fans must be at least 2 hours at 200°C - (3.2.1)… Electricity supply for emergency systems (lighting, communication, electrical supply etc.) must be fire resistant for 60 minutes or there must be an independent supply between two sources, fire resistant cable coverings, physical protection of cables	See also *(2.2.2)*
U.K.	[1]	See R11.	
U.S.A.	[1]	- (3-2.3.2) …conduits, raceways, ducts, boxes, cabinets, equipment enclosures, and their surface finish materials shall be capable of being subjected to temperatures up to 932°F (500°C) for 1 hour and shall not support combustion under the same temperature condition. Other materials, where encased in concrete or suitably protected, shall be acceptable. - (4-3.2) Emergency ventilation fans, their motors, and all related components shall resist at 250°C for a minimum of 1 hour. Fans rated in accordance with the ANSI/AMCA 210-85.	

- R14 Additional measures

Country	Ref.	Requirement	Comment
Italy		No reference.	
Switzerland		No reference.	
Germany		No reference.	
France		No reference.	
U.K.		No reference.	
U.S.A.		No reference.	

4.5 Emergency management

4.5.1 4.5M1 – Organisational measures

4.5.1.1 Role of the measure

To mitigate the consequences of an accident in a tunnel, plans are defined, published and made available for appropriate contingency measures and assigned responsibilities for managing emergency situations. The planning is lead by the tunnel's manager, in conjunction with all of those operating companies through the tunnel, managers of the adjacent infrastructure, local authorities and the emergency services. These organisational measures and emergency plans help to guarantee proper intervention and minimise time delays, although no plan can cover the multitude of possible occurrences.

4.5.1.2 Synthesis – comments

Analysis of the regulations reveal that the Italian, German, Swiss and Austrian regulations make specific reference to emergency plans and the way they must be elaborated. Among these regulations, those from Italy have a more limited field of application (only for tunnels longer than 3.000m), whereas the other countries extend their plans to include all tunnels on their respective networks. All of the regulations emphasise the necessity of periodic drills, though frequency of these differs from country to country. The French and U.K. regulations do not specifically refer to the development of emergency plans.

4.5.1.3 Comparison tables

- M11 Safety plans

Country	Ref.	Requirement	Comment
Italy	[1,7]	Starting from December 2001, emergency plans must be provided for all tunnels longer than 3000m. These plans must: • be drawn up with local competent authorities; • take into account possible risk scenarios; • provide for periodic drills.	
Switzerland	[!,3]	According to the "Ordinance on Protection Against Major Accidents" for every railway line and also for every tunnel, of whatever length, a contingency plan must be drawn up in conjunction with the emergency services and periodic exercises, on the basis of this plan, have to be carried out.	[!] Information obtained directly from railway network.
Germany	[1,4]	The infrastructure manager is required to draw up, with the rescue services, an operational alarm and hazard prevention plan, including the fire brigade plans pursuant to DIN 14095, for every tunnel. The plans are to be coordinated with the rural districts or urban communities and made available to these. Exercises with the rescue services must be carried out before a tunnel is put into service and at intervals of no more three years thereafter.	
France		No reference.	
U.K.		No reference.	
U.S.A.	[1]	- (7-8.1) (7-8-2) Emergency procedure plans are anticipated and shall clearly delineate the authority or participating agency that is in command and responsible for the overseeing, correction, or alleviation of the emergency. - (7-10.2) Exercises and drills …at least twice per year to prepare the authority and participating agency personnel for emergencies.	

UNECE	[10]	(C.4.13)….. regular maintenance of emergency and rescue plans is recommended as a standard safety measure. …...It is recommended that emergency service planning shall be developed during the tunnel's planning phase. …. it is recommended that the several organizations not only prepare together and regularly review their plans but also exercise jointly in various scenario situations.	
Other Austria		A plan for warning and danger must be drawn up for all tunnels, in order to foster collaboration among local organisations. This plan establishes: • the access ways for rescue teams on foot or by other means of transport; • the forces (teams) to be mobilised and the equipment they require; • the provisions for the infrastructure manager to make transport by rail available; • the methods used to give a shuttle service between the accident scene and tunnel entrances; • training methods.	

5 APPENDIX 1: TABLES OF CONTENTS OF NATIONAL GUIDELINES TRANSLATED INTO ENGLISH

5.1 *Italy (I)*

5.1.1 "Linee guida per il miglioramento della sicurezza nelle gallerie ferroviarie" (25.07.1997)

Ministry of the Interior, FS S.p.A., National Fire Brigade Corp.

Contents:

5.1.2 "Criteri progettuali per la realizzazione di piazzali di emergenza, le strade di accesso e le aree di atterraggio degli elicotteri ai fini della sicurezza delle gallerie ferroviarie" (Aug. 1998)

5.1.3 FS – RFI

Contents:

5.1.4 "Criteri progettuali per la realizzazione degli impianti idrico antincendio, elettrico e illuminazione, telecomunicazione, supervisione" (Apr. 2000)

5.1.5 FS – RFI

Contents:

5.1.6 "Linee guida per la realizzazione del piano generale di emergenza per lunghe gallerie ferroviarie" (Oct. 1998)

5.1.7 FS – RFI

Contents:

5.1.8 “Linee guida per l’elaborazione del piano interno di emergenza” (Jun. 2000)

5.1.9 FS-RFI

Contents:

5.1.10 "Linee guida per il tracciamento e la posa in opera di sistemi di supporto per cavo radiante nelle gallerie ferroviarie" (Apr. 2001)

FS – RFI

Contents:

1. General remarks
2. Fields of application
3. Typology of tunnels to be supplied with equipment
4. Support systems for radiant cables
   Fixing the cable with plastic supports and installation of the cable
   Fixing the cable with a rope fixed on the vault of the tunnel and installation of the cables
5. Fixation systems of the supply cables or fibreoptic cables
6. Papers to be supplied by the contractor

Attachments:
Attachment A: Principles of standard suspensions in railway tunnels
Attachment B: Minimum profile of obstacles for the pantograph L=1450 – chip 608.

5.1.11 RFI circular n° di/a1007/p/01/000562 del 04.06.2001: "Piano interno di emergenza per gallerie di lunghezza compresa tra i 5.000 m e 3.000 m"

FS-RFI

Contents:
Circular Department Infrastructure RFI
Arrangement of the internal emergency plans also for the tunnels with a length between 3 and 5 km, extending them to all the tunnels with a length > 3 km.

5.1.12 RFI circular n° rfi/tdr/a1007/p/01/000512 del 17.12.2001: "Standard di sicurezza per nuove gallerie ferroviarie"

FS-RFI

Contents:

Circular Department Investments RFI
Technical standard of new tunnels: with double track up to a length of 1000 m, tunnels with two tubes from 2000 m up, the designer has the possibility to choose the best solution for middle lengths.

5.2 Switzerland

5.2.1 "Raccomandazione comune delle autorità di vigilanza sulle ferrovie della Germania, dell'Austria e della Svizzera in merito alla sicurezza dei viaggiatori in gallerie ferroviarie molto lunghe" (24.09.1992)

Swiss Federal Office of Transport

Contents:

5.2.2 Prescrizioni svizzere sulla circolazione dei treni PCT

5.2.3 (R 300.1-.15, 14.12.2003)

UFT Federal Department of Transport

Contents:

The PCT contain the important security rules for the whole railway traffic, divided in the following parts:

.

5.2.4 Weisung I-AM-EB-31/00: "Sichereit in bestehenden Tunnels, Infrastrukturmassnhamen zur Erleichterung der Selbstrettung" (06.12.2000)

Swiss Federal Railways

Contents:

5.2.5 "Ausfuhrungsbestimmungen zur Eisenbahnverordnung von 16 Oktober 2002" (AB-EBV SR 742.141.11)

Contents:
Not available

5.2.6 "Verordnung vom 27 Februar 1991 über den Schutz vor Störfällen"

5.2.7 (Störfallverordnung, StFV SR 814.012)

Contents:

5.2.8 " Verordnung vom 23 November 1983 über Bau und Betrieb der Eisenbahnen“

(Eisenbahnverordnung EBV SR 742.141.1)

Contents:

5.3 GERMANY

5.3.1 Richtlinie "Anforderung des Brand- und Katastrophenschutzes an den Bau und Betrieb von Eisenbahntunneln" (01.07.1997).

Eisebahn-Bundesamt (EBA)

Contents:

5.3.2 "Leitfaden für den Brandschutz in Personenverkehrsanlagen der Eisenbahnen des Bundes und der Magnetschnellbahn" (Jan. 2001)

Eisenbahn-Bundesamt (EBA)

Contents:

5.3.3 "Anforderungen der DB station&service AG an den Brandschutz in Personenverkehrsanlagen" - Draft Version 15.03.2001

Deutsche Bahn AG (DB)

Contents:

5.3.4 " FIRE AND DISASTER PROTECTION IN RAILWAY TUNNELS" – (Mar. 2004)

Deutsche Bahn AG (DB)

Contents

List of illustrations:

5.4 FRANCE

5.4.1 "Instruction téchnique interministérielle rélative à la sécurité dans les tunnels ferroviaires" (08.07.1998)

Ministere de l'Equipement, des Transportes et du Logement

Contents :

5.4.2 "Décret relatif a la sécurité du reseau ferre national (30.05.2000)

Ministère de l'Equipement, des Transportes et du Logement

Contents:

5.5 NORWAY

5.5.1 "Substructure, regulations for new lines, tunnels – safety requirements" (01.01.2000)

Jernbaneverket

Contents :

5.6 AUSTRIA

5.6.1 Richtlinie "Bau und Betrieb von Neuen Eisenbahntunneln bei Haupt- und Nebenbahnen, Anforderungen des Brand- und Katastrophenschutzes" (1. Ausgabe 2000)

Österreichischer Bundesfeuerwehrverband

Contents

5.6.2 Eisbav: "Eisenbahn-Arbeitnehmerinnershutzverordnung" (Jun. 2001)

Verkehrs- Arbeitsinspektorat
Contents:

5.6.3 "Allgemeines Sicherheitskonzept Für Mittlere Tunnel" (Dec.1995).

5.6.4 OBB - Austrian Federal Ministry Of Transport - Eisenbahn – Hochleistungsstrecken AG

Contents:

5.6.5 "Allgemeines Sicherheitskonzept Für Mittlere Bestandestunnel" (Aug 1996)

5.6.6 OBB - Austrian Federal Ministry Of Transport - Eisenbahn – Hochleistungsstrecken AG

Contents

5.6.7 "Allgemeines Sicherheitskonzept Für lange Tunnel" (Dec 1995)

5.6.8 OBB - Austrian Federal Ministry Of Transport - Eisenbahn – Hochleistungsstrecken AG

Contents:

5.6.9 „Richtilinien Für das Entwerfen von Bahnanlagen - Hochleistungsstrecken (HL- Richtilinien)“

Contents

5.6.10 Handbuch " Feuerwehreinsatz im Gleisbereich“

5.6.11 ÖBB - ÖBFV

Contents
Not available

5.7 UNITED KINGDOM

5.7.1 "Railway Safety Principles And Guidance, Part 2, Section A, Guidance On The Infrastructure - Chapter 5: Tunnels" (Mar. 2002)

UK Health And Safety Executive
Contents:

5.7.2 "System Safety Requirements for New and Re-opened Tunnels"

Railway Group Standard – Railway Safety GC/RT 5114 Draft 3f December 2002
Contents:

5.7.3 "Guidance on System Safety Requirements in New and Re-opened Tunnels"Railway Group Guidance Note – Railway Safety GC/GN 5614 Draft 2f December 2002

Contents:

5.8 Spain

5.8.1 "Instrucción obras subterráneas" (01.12.1998)

Renfe – Ministirio De Fomento

Contents :

5.8.2 "Medidas De Seguridad En Nuevos Tuneles Ferroviarios" (Apr. 2000)
Renfe

Contents:

5.9 The Netherlands

5.9.1 "The Dutch Vision On Safety In Road And Rail Tunnels (Draft)" (2003)
Ministries of Transports And Of Inland Affairs.

Contents:

9. Contingency planning
10. PR, education and training

5.10 SWEDEN

5.10.1 "Säkerhet I Järnvägstunnlar, Ambitionsnivå Och Värderingsmetodik, Handbok BVH 585.30" (01.09.1997) - Zusammenfassung Auf Englisch

Banverket
Contents:
The handbook offers a basis for objective-guided design by providing procedure for risk evaluations, establishing criteria on the tolerable level of safety and suggesting additional safety measures.

5.11 Finland

5.11.1 "Technical Regulation And Guidelines For Railways: Railway Tunnels" (Oct. 2002)

RAMO – Finnish Rail Administration Board

Contents:

18.7 Technical Systems
18.71 Extensiveness of the systems
18.72 Electrical systems
18.721 Transmission and distribution of electric energy
18.722 Main and group distribution centres
18.723 Back-up power supply
18.724 Cable routes
18.73 HVAC (heating, plumbing and air-conditioning) systems
18.731 Ventilation
18.732 Sewerage
18.73 3 HVAC monitoring system
18.74 Lighting
18.741 Normal lighting
18.742 Back-up lighting
18.75 Safety and monitoring systems
18.751 Fire annunciator systems
18.752 Fire extinguishing systems
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18.754 Aerial systems
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5.12 USA

5.12.1 "NFPA 130 Standards For Fixed Guideway Transit And Passenger Rail Systems 2003 Edition" (May 2003) - National Fire Association

Contents:

6 APPENDIX 2: TABLES OF CONTENTS OF OTHER REFERENCE DOCUMENTS (TRANSLATED INTO ENGLISH).

6.1 UIC

6.1.1 UIC Codex 779-9 "Safety In Railway Tunnels – Recommendations for safety measures final report" (24/09/2002)

Contents:

6.1.2. "MEASURES TO LIMIT AND REDUCE THE RISK OF ACCIDENTS IN UNDERGROUND RAILWAY INSTALLATIONS WITH PARTICULAR REFERENCE TO THE RISK OF FIRE AND THE TRANSPORT OF DANGEROUS GOODS" (JUNE 1991)

Contents:

6.2 U.N.

6.2.1 Report Of The Ad Hoc Multidisciplinary Group Of Experts On Safety In Tunnels (Rail). (Jul. 2003)

Contents:

6.2.2 “Questionnaire on safety in railway tunnels – Replays by different Countries”

UN Informal Documents No. 14-15-16 (Nov. 2002)

Contents:
This summary contains the tabular presentation of the replies to the questionnaire on Safety in Railway Tunnels received by the secretariat from member Governments.

6.2.3 “A Summary of Accidents in Railway Tunnels”

UN Informal Document No. 8 (Nov. 2002)

Contents:
The document contains a summary of accident reports. Accidents have been recorded in chronological order and each page identified by both page number and the country in which the accident occurred together with a sequential identification number for that country

6.2.4 “Railway tunnels in Europe and North America”

UN Secretariat – Informal Document No. 7 (May 2002)

Contents:
The document contains the list of railway tunnels (longer than 1.000m), compiled by UN Secretariat from various national and international sources. The list is intended to serve as a reference inventory a long railway tunnels in Europe and North America.

6.2.5 “Vehicles Fire And Fire Safety In Tunnels”

UN Secretariat – Informal Document No. 9 (Sept. 2002)
Centre For Fire Safety In Transport, UK

Contents:
This paper is a copy of the article “Vehicle fires and fire safety in tunnels” by Martin Shipp, Centre for Fire Safety in Transport, Building Research Establishment Ltd. United Kingdom, published in the “Tunnel Management International”, Vol. 5, No.3, 2002.

- Abstract
- Introduction
- Statistics
- Fire Severity
- Railway Standards and Codes
- Road Vehicle Standards and Codes
- Ignition Scenarios

- Fire Growth, Fire Development and Reaction to Fire
- Fire Resistance and Compartmentation
- Detection Systems
- Alarm and Warning Systems
- Smoke Control Systems
- Means of Escape, Egress Provisions (Doors, Windows or Hatches), Places of Relative Safety and Places of Safety
- Fire Suppression and Availability of Fire Fighting Media, Fire Fighting and Fire Service Response
- Interaction with the Infrastructure
- Possible Risk Reduction Measures
- Conclusions
- References
- Acknowledgements

6.2.6 Peut-On Garantir La Securite Des Voyageurs Dans Les Longs Tunnels Ferroviaires ? UN Secretariat – Informal Document No. 1 (Jun 2003)

Alptransit Gothard Sa

Contents:
The aim of the document is to give an answer according to the safety measures planned for the base tunnel of Saint Gotthard that will be the longest railway tunnel in the world with 57 km.

6.2.7 "The Safety of the Swiss railway tunnels – Analysis of the federal office of transport" UN Informal Document N° 2 (Jun 2003)

UN – SECRETARIAT

Content:
New breakdown of Swiss rail tunnels
Railways – a safe mode of transport
Measures to promote autonomous rescue
Rapid assistance is decisive
Poor access for road vehicles
Quality of rolling stock: a safety factor
Raising the safety standard to the safety level of new tunnels

6.2.8 A "Protection against fire and other catastrophic events in railway tunnels"

UN Informal Document N° 13 (Jun 2003)

Contents :

6.2.9 A "Fire Protection in transport tunnels" (Germany)

UN Informal Document N° 13 (Jun 2003)

Contents:

6.3 AEIF – European Association for Railway Interoperability

6.3.1 Mandate for CR second priority TSIs (version04)

Contents:

6.4 SBB-CFF-FFS

6.4.1 "Nutzungsanforderungen an neue Eisenbahntunne“ (Mar 2001)

Use requirements for new railway tunnels“ (March 2001)

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7.3 Additional technical spaces
7.4 Carriageway (Fb)
7.5 Train power supply, power line (EA-FI)
7.6 Cable installations (EA-K)
7.7 Low tension, electromechanical equipment (6VN)
7.8 Safety installations (SA-S)
7.9 Telecommunications installations (TG)
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6.10.3 Beveiligingsconcept Hsl-Zuid, Teile A, B1, B2, B3

Hsl-Zuid
Contents:
Not available

6.10.4 "High Speed Line South: Safety Concept- Green Heart Tunnel" UN Informal Document N° 10 (Jun 2003)

Government of Netherlands
Contents:

6.11 UFT (SWISS FEDERAL OFFICE OF TRANSPORT)

6.11.1 Rapport Final Sur La Securite Dan Le Tunnels Ferroviaires Suisses

Final Report about the Security of the Swiss Railway Tunnels

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